

Visualization of endogenous G proteins on endosomes and other organelles

Reviewed Preprint

v2 • September 25, 2024

Revised by authors

Reviewed Preprint

v1 • May 2, 2024

Wonjo Jang, Kanishka Senarath, Gavin Feinberg, Sumin Lu, Nevin A Lambert 

Department of Pharmacology and Toxicology, Medical College of Georgia, Augusta University, Augusta, GA, 30912, USA

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

Abstract

Classical G protein-coupled receptor (GPCR) signaling takes place in response to extracellular stimuli and involves receptors and heterotrimeric G proteins located at the plasma membrane. It has recently been established that GPCR signaling can also take place from intracellular membrane compartments, including endosomes that contain internalized receptors and ligands. While the mechanisms of GPCR endocytosis are well understood, it is not clear how well internalized receptors are supplied with G proteins. To address this gap we use gene editing, confocal microscopy, and bioluminescence resonance energy transfer to study the distribution and trafficking of endogenous G proteins. We show here that constitutive endocytosis is sufficient to supply newly internalized endocytic vesicles with 20-30% of the G protein density found at the plasma membrane. We find that G proteins are present on early, late, and recycling endosomes, are abundant on lysosomes, but are virtually undetectable on the endoplasmic reticulum, mitochondria, and the medial Golgi apparatus. Receptor activation does not change heterotrimer abundance on endosomes. Our findings provide a subcellular map of endogenous G protein distribution, suggest that G proteins may be partially excluded from nascent endocytic vesicles, and are likely to have implications for GPCR signaling from endosomes and other intracellular compartments.

eLife assessment

This **important** study investigates the intracellular localization patterns of G proteins involved in GPCR signaling, presenting **compelling** evidence for their preference for plasma and lysosomal membranes over endosomal, endoplasmic reticulum, and Golgi membranes. This discovery has significant implications for understanding GPCR action and signaling from intracellular locations. This research will interest cell biologists studying protein trafficking and pharmacologists exploring localized signaling phenomena.

<https://doi.org/10.7554/eLife.97033.2.sa4>

Introduction

Heterotrimeric G proteins transduce a vast number of important physiological signals (Gilman, 1987 [↗](#)), most often in response to activation by G protein-coupled receptors (GPCRs) (Pierce et al., 2002 [↗](#)). Canonical G protein signaling occurs when a cell surface GPCR is activated by an extracellular ligand, which in turn promotes activation of plasma membrane G protein heterotrimers and downstream effectors. Recently, it has become clear that GPCRs can also signal from intracellular compartments (Calebiro et al., 2010 [↗](#); Eichel & von Zastrow, 2018 [↗](#)), most notably endosomes and the Golgi apparatus (Calebiro et al., 2009 [↗](#); Ferrandon et al., 2009 [↗](#); Irannejad et al., 2017 [↗](#); Irannejad et al., 2013 [↗](#); Mullershausen et al., 2009 [↗](#)). Signaling from endosomes is often a continuation of signaling that starts at the plasma membrane and persists as (or resumes after) active receptors are endocytosed (Tsvetanova et al., 2015 [↗](#)). Much is known about the machinery responsible for GPCR internalization, and also about the trafficking itineraries of specific receptors after endocytosis. Some receptors are efficiently sorted for recycling and are returned to the plasma membrane, whereas other receptors are rapidly degraded (Hanyaloglu & von Zastrow, 2008 [↗](#)). It is also known that at least one isoform of the G protein effector adenylyl cyclase is actively internalized (Lazar et al., 2020 [↗](#)).

In contrast, much less is known about how G protein heterotrimers traffic from the plasma membrane through intracellular compartments (Wedegaertner, 2012 [↗](#)). It is not known how efficiently heterotrimers are loaded onto endocytic vesicles at the plasma membrane, how receptor activation might change this process, or what the fate of G proteins might be after endocytosis. Activation at the plasma membrane promotes heterotrimer dissociation, and the resulting loss of membrane avidity allows G $\beta\gamma$ dimers and some G α subunits (most notably G α_s) to translocate through the cytosol to sample intracellular membranes (Akgoz et al., 2004 [↗](#); Hynes et al., 2004 [↗](#); Ransnäs et al., 1989 [↗](#); Slepak & Hurley, 2008 [↗](#); Wedegaertner et al., 1996 [↗](#)). However, these processes reverse quickly when activation ceases (Akgoz et al., 2004 [↗](#); Martin & Lambert, 2016 [↗](#)), meaning that activation-dependent translocation of free G α subunits and G $\beta\gamma$ dimers would be an inefficient mechanism to deliver inactive heterotrimers to intracellular membranes. G proteins have been detected on the surface of endosomes and other intracellular compartments using a variety of approaches (Hewavitharana & Wedegaertner, 2012 [↗](#); Irannejad et al., 2013 [↗](#); Scarselli & Donaldson, 2009 [↗](#); Wedegaertner, 2012 [↗](#)). However, there has been no quantitative comparison of G protein distribution across identified subcellular compartments. Studies of endogenous G proteins are limited by the availability of well-validated antibodies suitable for immunostaining, and overexpression of tagged G protein subunits may lead to aberrant localization.

Here we study the subcellular distribution of endogenous heterotrimeric G proteins in cultured cells using CRISPR-mediated gene editing, confocal imaging, and bioluminescence resonance energy transfer (BRET). We find that G proteins are abundant on membrane compartments that are functionally continuous with the plasma membrane, including early, late, and recycling endosomes. However, heterotrimer density on endocytic membranes is lower than on the plasma membrane, suggesting that G protein endocytosis is inefficient. Endocytic trafficking of G proteins is not regulated by GPCRs. Our findings are likely to have implications for GPCR signaling from endosomes, as internalized receptors are concentrated in G protein-deficient compartments.

Results

To study the localization of endogenous G proteins we used gene editing to attach small peptide tags to the amino terminus of G β_1 subunits (*GNB1*) in HEK 293 cells. We chose this subunit because it is the most abundant G β subunit in this cell type (Cho et al., 2022 [\[1\]](#)), it can associate with any type of G α or G γ subunit (Hillenbrand et al., 2015 [\[2\]](#)), and it can be labeled at this position without disrupting heterotrimer formation or function. For bioluminescence experiments we added the HiBit tag (Schwinn et al., 2018 [\[3\]](#)) and isolated clonal “HiBit- β_1 ” cell lines. An advantage of this approach over adding a full-length Nanoluc luciferase is that it requires coexpression of LgBit to produce a complemented luciferase. This limits luminescence to cotransfected cells and thus eliminates background from untransfected cells. For imaging experiments we added a tandem tag that included the 11th beta strand of mNeonGreen2 (mNG2(11)) (Feng et al., 2017 [\[4\]](#)) and HiBit in cells constitutively expressing mNG2(1-10) and isolated “mNG- β_1 ” cell lines (**Figure 1A** [\[5\]](#)). Amplicon sequencing verified that cell lines had correctly edited *GNB1* genes and SDS-PAGE revealed single proteins with apparent molecular weights consistent with edited G β_1 subunits (**Figure 1B** [\[6\]](#)). BRET assays demonstrated that tagged subunits in HiBit- β_1 and mNG- β_1 cell lines formed functional heterotrimers with endogenous G α and G γ subunits. For example, we observed agonist-induced BRET between the D2 dopamine receptor and mNG- β_1 , an interaction that requires association with endogenous G α subunits (**Figure 1C** [\[7\]](#)). Similarly, we observed BRET between HiBit- β_1 and the free G $\beta\gamma$ sensor memGRKct-Venus after activation of receptors that couple G $\alpha_{i/o}$, G α_s , and G α_q heterotrimers, indicating that HiBit- β_1 associated with endogenous G α subunits from these three families (**Figure 1D** [\[8\]](#)). We also found that cyclic AMP accumulation in response to stimulation of endogenous β adrenergic receptors was similar in edited cell lines and their unedited parent lines (Figure 1 – figure supplement 1). Endogenous G α and G β subunits are expressed at approximately a 1:1 ratio, and G β subunits are tightly associated with G γ and inactive G α subunits (Cho et al., 2022 [\[1\]](#); Gilman, 1987 [\[9\]](#); Krumins & Gilman, 2006 [\[10\]](#)). Moreover, proteins that bind to free G $\beta\gamma$ dimers are found in the cytosol of unstimulated HEK 293 cells, suggesting at most only a small population of free G $\beta\gamma$ in these cells (Barak et al., 1999 [\[11\]](#)). Therefore, we assume that almost all mNG- β_1 and HiBit- β_1 subunits in unstimulated cells are part of heterotrimers.

Endogenous G proteins primarily associate with the plasma membrane and endolysosomes

Confocal imaging of mNG- β_1 cells revealed the expected bright fluorescence at the plasma membrane. Most cells also contained pleiomorphic intracellular structures and dim cytosolic fluorescence that was sufficient to suggest relative exclusion of mNG- β_1 from the nucleus. Especially notable were clusters of large vesicular structures located at the cell periphery which were later identified as lysosomes (**Figure 2A** [\[12\]](#); see below). Large intracellular organelles such as the endoplasmic reticulum, mitochondria, and Golgi apparatus were not evident. Unsurprisingly, our images are quite similar to those made as part of previous study that labeled G β_1 subunits with mNG2 (Cho et al., 2022 [\[1\]](#)).

To identify the intracellular membrane compartments with mNG- β_1 fluorescence we coexpressed a series of organelle markers tagged with red fluorescent proteins. Markers of the endoplasmic reticulum, mitochondria and medial Golgi apparatus indicated that these large compartments were virtually devoid of mNG- β_1 fluorescence (**Figure 2B-D** [\[13\]](#), Figure 2 – figure supplements 1-3). In some cells an indistinct region of mNG- β_1 fluorescence was interleaved with leaflets of the Golgi apparatus, but line profiles suggested that this was a distinct structure (**Figure 2D** [\[14\]](#), Figure 2 – figure supplement 3), most likely the perinuclear recycling compartment (see below).

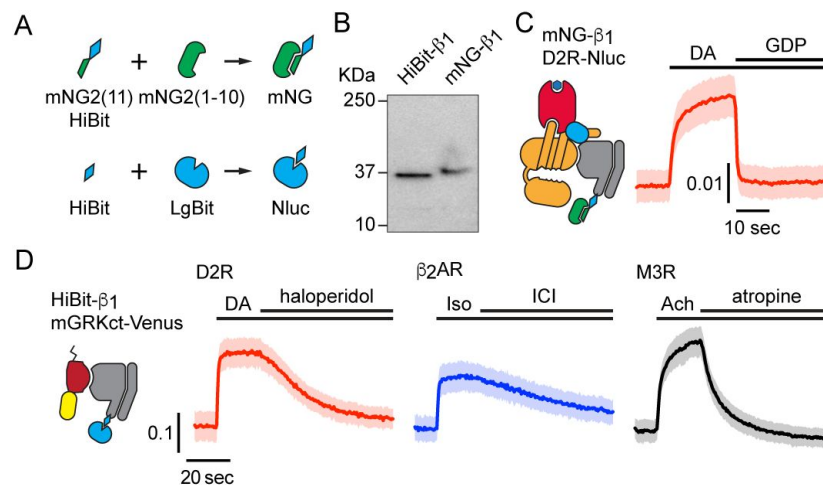


Figure 1 with 1 supplement

Validation of mNG-β₁ and HiBit-β₁ cells.

(A) Cartoon showing the peptide tag complementation systems used to label endogenous Gβ₁ subunits. (B) SDS-PAGE of HiBit-β₁ and mNG-β₁ cell lysates; the predicted molecular weights of the edited gene products are 38.9 and 41.1 kilodaltons (KDa), respectively; representative of 3 independent experiments. (C) In permeabilized nucleotide-depleted cells BRET between dopamine D2R-Nluc receptors and mNG-β₁-containing heterotrimers increases in response to dopamine (DA; 100 μM) and reverses after addition of GDP (100 μM); mean ± 95% CI; *n*=27 replicates from 2 independent experiments. (D) In intact cells BRET between HiBit-β₁ and the Gβγ sensor memGRKct-Venus increases after stimulation of D2R dopamine, β₂AR adrenergic, or M3R acetylcholine receptors with DA (100 μM), isoproterenol (Iso; 10 μM) and acetylcholine (Ach; 100 μM), respectively. Signals reversed when receptors were blocked with haloperidol (10 μM), ICI 118551 (10 μM) or atropine (10 μM); mean ± 95% CI; *n*=16 replicates from 4 independent experiments.

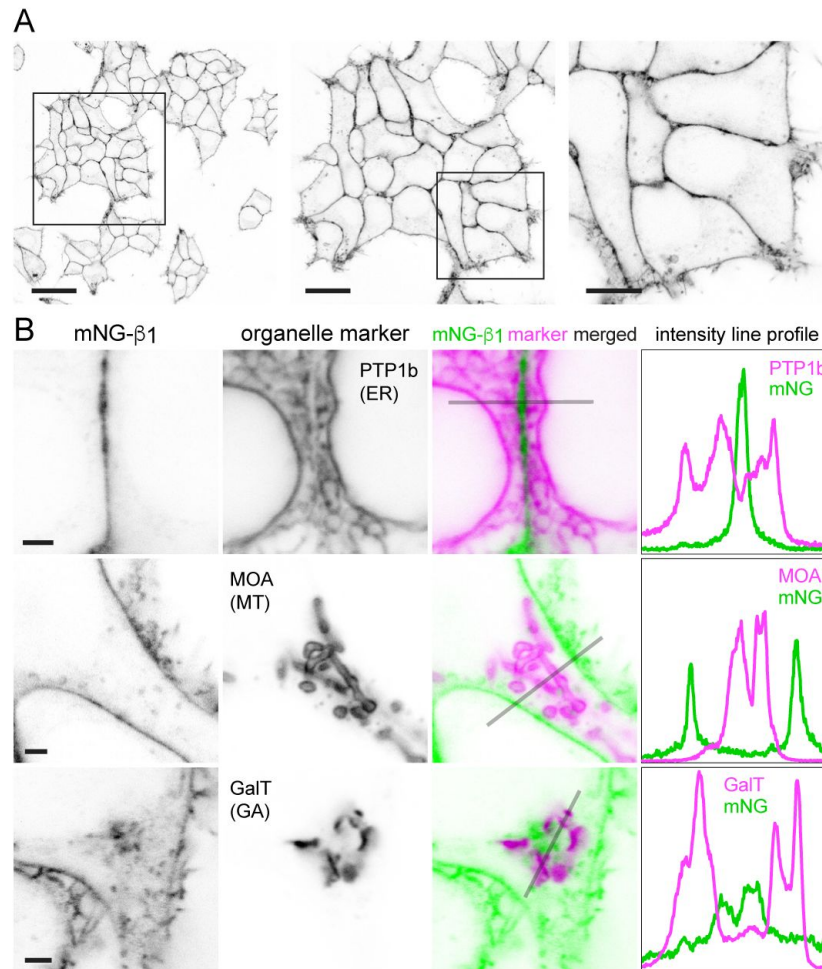


Figure 2 with 3 supplements

Endogenous G proteins are abundant on the plasma membrane but not large organelles.

(A) A single field of view of mNG-β₁ cells at three magnifications; scale bars are 40 μm, 20 μm and 10 μm. (B) mNG-β₁ does not colocalize with expressed markers of the endoplasmic reticulum (ER; PTP1b), mitochondria (MT; MOA) or medial Golgi apparatus (GA; GalT); intensity line profiles depict absolute fluorescence intensity in each channel; scale bars are 2 μm.

In contrast, mNG- β_1 clearly colocalized with the marker FYVE, which binds to phosphatidylinositol-3-phosphate (PI3P) on the surface of endosomes (**Figure 3A**, B, Figure 3 – figure supplement 1). However, mNG- β_1 fluorescence was not detected on every FYVE-positive vesicle and when present was much less intense than fluorescence of adjacent segments of the plasma membrane (**Figure 3B**). The median signal-to-background ratio for FYVE-positive structures was less than one-fourth that of the plasma membrane (**Figure 3C**). FYVE domains primarily localize to early endosomes (Hammond & Balla, 2015), so it was not surprising that similar colocalization of mNG- β_1 was observed with the early endosome marker rab5a. As was the case with FYVE, mNG- β_1 was detectable in some rab5a-positive vesicles but not others and was not as intense as the nearby plasma membrane (**Figure 3A**, C, Figure 3 – figure supplement 2). In order to determine the fate of G proteins after endocytosis we then examined mNG- β_1 colocalization with markers of recycling and late endosomes (Stenmark, 2009). Dim mNG- β_1 fluorescence was detected on indistinct rab11a-positive structures clustered diffusely in the vicinity of the nucleus (**Figure 3A**, Figure 3 – figure supplement 3), which we presumptively identified as the perinuclear recycling compartment (PNRC). Similarly, mNG- β_1 colocalized extensively with vesicles labeled with rab7a, a marker of late endosomes (**Figure 3A**, Figure 3 – figure supplement 4). Notably, mNG- β_1 fluorescence was more intense on rab7a-positive late endosomes than on FYVE- or rab5a-positive early endosomes (**Figure 3C**). The presence of mNG- β_1 on late endosomes suggested that some G proteins may be degraded by lysosomes. Accordingly, mNG- β_1 strongly colocalized with lysosomes marked with LysoView 633 (**Figure 3A**, Figure 3 – figure supplement 5), long-term incubation with fluorescent dextran (Figure 3 – figure supplement 5), or the lysosome marker LAMP1 (Figure 3 – figure supplement 6). While some mNG fluorescence was detected in the lumen of lysosomes, much of the signal remained on the cytosolic surface of these structures, and in many instances the intensity of mNG- β_1 fluorescence on lysosomes was similar to that of the nearby plasma membrane (**Figure 3A**). We also detected robust mNG- β_1 signals on structures labeled with TGNP (Figure 3 – figure supplement 7), a marker of the trans-Golgi network. These imaging results suggest that G proteins are likely to undergo endocytosis and enter both recycling and degradative pathways and may become more concentrated as late endosomes mature.

As an alternative approach we performed bystander BRET experiments to map the subcellular localization of endogenous HiBit- β_1 . This approach provides an unbiased index of membrane protein colocalization from large populations of cells and has the additional advantage of very high sensitivity (Lan et al., 2012). We expressed LgBit and a series of inert Venus-tagged membrane markers in HiBit- β_1 cells and observed large bystander signals at the plasma membrane, smaller bystander signals at endosomes, and very small bystander signals at the endoplasmic reticulum and mitochondria (**Figure 3D**). Although bystander BRET signals cannot be directly compared between different compartments, these results are generally consistent with what we observed using confocal imaging and confirm the presence of G proteins on multiple endosomal compartments.

Constitutive G protein endocytosis is inefficient

That mNG- β_1 fluorescence was less intense on endosomes than the plasma membrane suggested that G protein density may be lower on the surface of endosomes than on the plasma membrane. However, differences in fluorescence intensity could also be due to differences in the amount of membrane surface area sampled in the imaging volume. Likewise, differences in bystander BRET between compartments could be due to differences in several factors, including expression and efficiency of compartment-specific BRET acceptors. Therefore, we devised a co-labeling protocol that allowed us to compare mNG- β_1 fluorescence to the amount of newly internalized membrane imaged at endocytic vesicles, and to simultaneously make the same measurements at the plasma membrane. To stain both the plasma membrane as well as newly formed endocytic vesicle membrane we exposed live cells to the styryl dye FM4-64, which rapidly and reversibly partitions

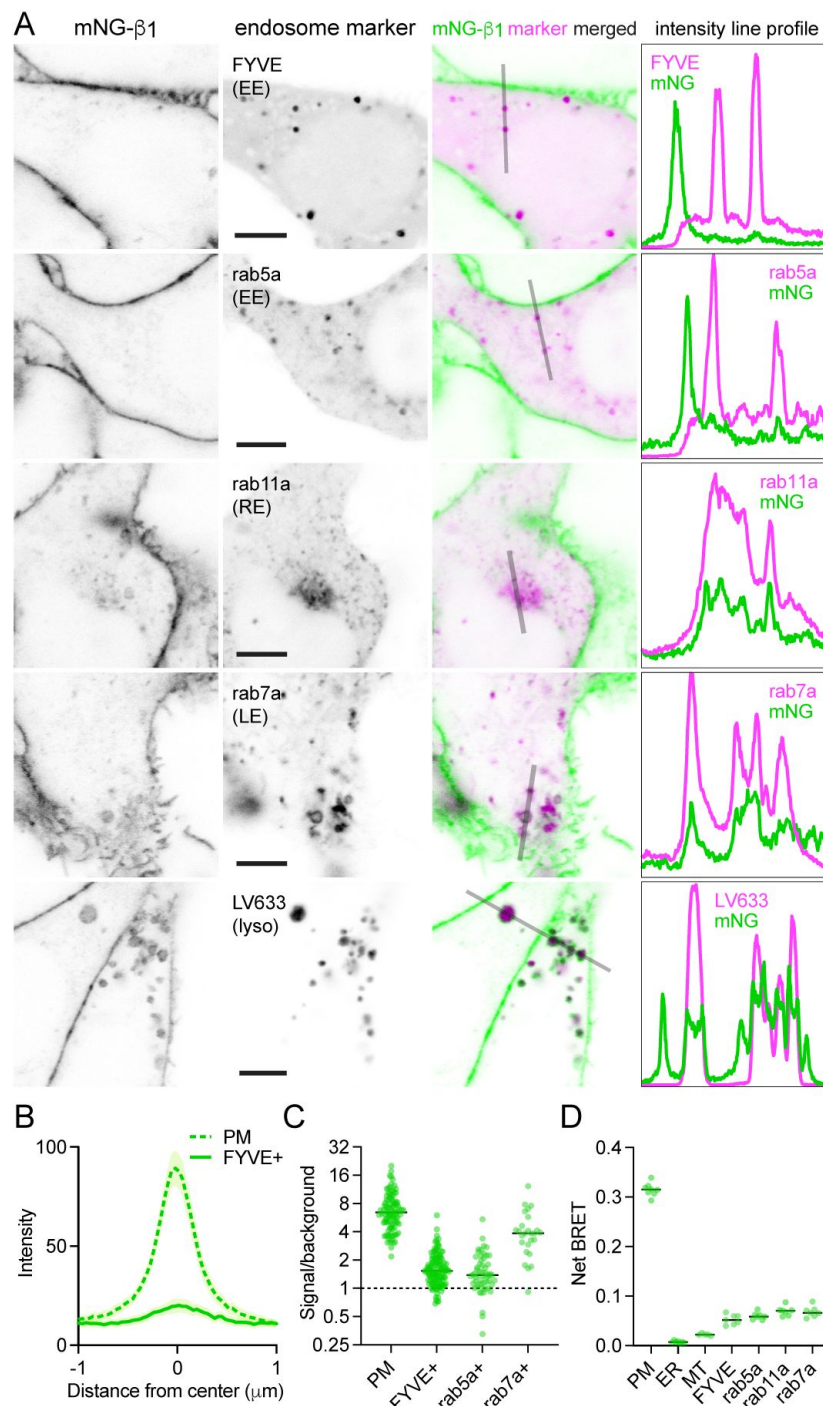


Figure 3 with 7 supplements

Endogenous G proteins colocalize with markers of endosomes and lysosomes.

(A) mNG- β_1 colocalizes with expressed markers of early endosomes (EE; FYVE and rab5a), recycling endosomes (RE; rab11a), late endosomes (LE; rab7a) and lysosomes (lyso; LysoView 633); intensity line profiles depict absolute fluorescence intensity in each channel; scale bars are $5\mu\text{m}$. (B) Mean mNG- β_1 fluorescence intensity line profiles drawn across the plasma membrane (PM) and FYVE-positive vesicles; mean $\pm$ 95% CI; $n=40$ vesicles/cells. (C) mNG- β_1 signal/background ratios for regions of interest surrounding the plasma membrane (PM; $n=99$), FYVE-positive ($n=125$) and rab5a-positive ($n=56$) early endosomes, and rab7a-positive ($n=26$) late endosomes; horizontal lines represent the median. (D) Bystander net BRET signals between HiBit- β_1 and Venus-tagged markers of the plasma membrane (PM), endoplasmic reticulum (ER), mitochondria (MT), early endosomes (FYVE and rab5a), recycling endosomes (rab11a) and late endosomes (rab7a); horizontal lines represent the median; $n=5$ -7 independent experiments.

into (but does not cross) membranes and is only fluorescent in a hydrophobic environment (Betz et al., 1996). When cells are exposed to FM4-64 at physiological temperatures the plasma membrane is stained immediately, and this is followed over the course of several minutes by the appearance of intracellular vesicles that have trapped the dye (Figure 4A, Figure 4 – figure supplement 1). As an orthogonal approach we stained cells with CellMask Deep Red, a lipophilic dye that permanently stains the plasma membrane and therefore is incorporated into endocytic vesicles. Both dyes are expected to produce fluorescence signals proportional to the surface area of the membrane sampled by the imaging method, allowing us to normalize the fluorescence of individual vesicles to the nearby plasma membrane. We reasoned that if G proteins are passively incorporated into endocytic vesicles without any enrichment or exclusion, then mNG- β_1 fluorescence in each vesicle should have the same intensity relative to the plasma membrane as lipophilic dyes. After staining cells and allowing 15 minutes for constitutive endocytosis we found that FM4-64 and CellMask dyes reported similar amounts of membrane surface area in endocytic vesicles (Figure 4A, B); in both cases peak vesicle intensity was on average similar to the intensity of the plasma membrane (Figure 4C). In contrast, peak mNG- β_1 fluorescence on the same endocytic vesicles was much less intense than the plasma membrane (Figure 4A-C). There was considerable variability between individual vesicles, such that some vesicles contained no detectable mNG- β_1 fluorescence (Figure 4A, Figure 4 – figure supplement 1). Using FM4-64 fluorescence as a standard for membrane surface area we calculated that mNG- β_1 density on FM4-64-positive vesicles was $20 \pm 8\%$ (mean $\pm$ 95% CI; $n=45$) of the nearby plasma membrane. This result confirms that heterotrimeric G proteins are present on newly internalized membrane but also suggests that G proteins are partially excluded from endocytic vesicles. To test if other peripheral membrane proteins are similarly depleted from endocytic vesicles, we performed analogous experiments by overexpressing mNG bearing the C-terminal membrane anchor of HRas (mNG-HRas ct). We found that mNG-HRas ct was also less abundant on FM4-64-positive endocytic vesicles than on the plasma membrane, although not to the same extent as mNG- β_1 (Figure 4 – figure supplement 2); mNG-HRas ct density on FM4-64-positive vesicles was $64 \pm 17\%$ (mean $\pm$ 95% CI; $n=78$) of the nearby plasma membrane.

Receptor activation does not change G protein endocytosis

The above results suggested that constitutive endocytosis of heterotrimeric G proteins is inefficient. However, it is possible that GPCR and G protein activation could change how G proteins are loaded onto endocytic vesicles. To test this possibility, we performed similar imaging experiments with mNG- β_1 cells transfected with SNAP-tagged β_2 adrenergic receptors (SNAPf- β_2 AR). This receptor is often used as a model of activity-dependent GPCR internalization (Benovic et al., 1988; von Zastrow & Kobilka, 1992) and has been shown to activate G proteins on endosomes (Bowman et al., 2016; Irannejad et al., 2013). We labeled SNAPf- β_2 AR with a membrane-impermeant SNAP ligand (AF 647) at room temperature to prevent constitutive endocytosis, then incubated cells with FM4-64 and the agonist isoproterenol for 15 minutes at physiological temperature to promote receptor endocytosis. Confocal imaging after agonist washout revealed numerous intracellular vesicles with intense AF 647 fluorescence, consistent with robust receptor internalization (Figure 5A). Normalization and comparison to FM4-64 fluorescence indicated that SNAPf- β_2 AR was enriched approximately three-fold on endocytic vesicles compared to the nearby plasma membrane (Figure 5B, C), consistent with active recruitment of active receptors to clathrin-coated pits and endocytic vesicles. In contrast, mNG- β_1 fluorescence in the same vesicles was again lower than expected given the amount of membrane imaged in each vesicle (Figure 5B, C). Once again there was considerable variability between individual endocytic vesicles (Figure 5A, D). Using FM4-64 fluorescence as a standard for membrane surface area we calculated that mNG- β_1 density on receptor-containing vesicles was $28 \pm 8\%$ (mean $\pm$ 95% CI; $n=91$) of the nearby plasma membrane. Although this density was higher

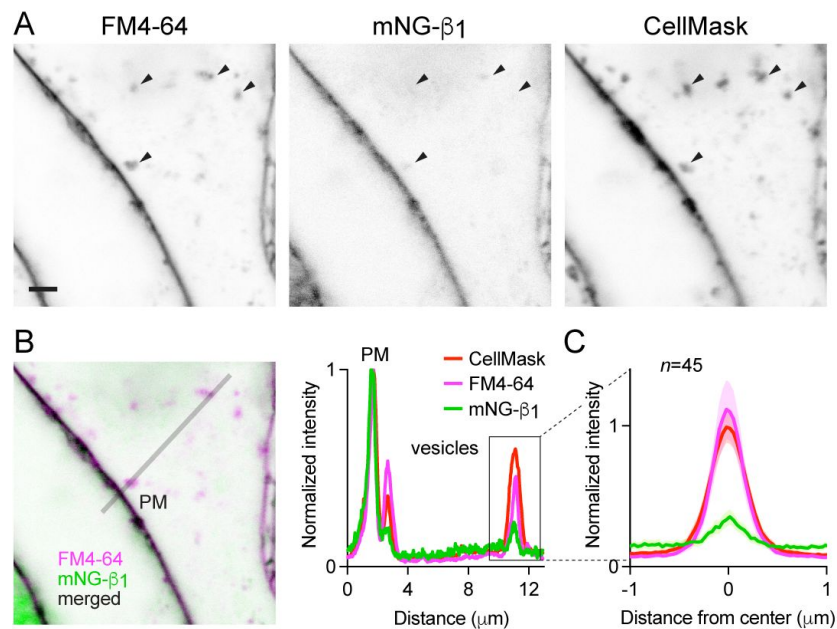


Figure 4 with 2 supplements

Constitutive G protein endocytosis is inefficient.

(A) mNG-β₁ colocalizes with newly internalized endocytic vesicles labeled with FM4-64 and CellMask Deep Red (arrowheads); scale bar is 2 μm. (B) A fluorescence intensity line profile for mNG-β₁, FM4-64 and CellMask normalized to the peak value of each label at the plasma membrane (PM). (C) Mean mNG-β₁, FM4-64 and CellMask fluorescence intensity line profiles drawn across vesicles, normalized to fluorescence intensity at the plasma membrane for each label; mean ± 95% CI; $n=45$ vesicles/cells.

than that calculated for vesicles formed by constitutive endocytosis, the difference did not reach significance (**Figure 5D** [↗](#)). These results demonstrate that activation-dependent internalization of β_2 adrenergic receptors does not significantly promote or prevent loading of G proteins onto endocytic vesicles.

These findings suggested that β_2 adrenergic receptor activation should have no impact on the abundance of heterotrimeric G proteins on endosomes. To test this idea for this and other receptors, we performed bystander BRET experiments with HiBit- β_1 cells transiently expressing β_2 adrenergic, D2 dopamine or M3 muscarinic receptors to activate G_s , $G_{i/o}$ and $G_{q/11}$ heterotrimers, respectively. With overexpressed G proteins free $G\beta\gamma$ dimers translocate from the plasma membrane and sample intracellular membrane compartments when dissociated from $G\alpha$ subunits. Consistent with this, we observed small decreases in bystander BRET at the plasma membrane and small increases in bystander BRET at intracellular compartments during activation of GPCRs, suggesting that endogenous $G\beta\gamma$ subunits undergo similar translocation (Figure 5 – figure supplement 1). Notably, these changes occurred at room temperature, suggesting that endocytosis was not involved, and developed over the course of minutes. The latter observation and the small magnitude of agonist-induced changes are both consistent with expression of primarily slowly-translocating endogenous $G\gamma$ subtypes in HEK 293 cells. Moreover, as shown previously for overexpressed $G\beta\gamma$, the changes we observed with endogenous $G\beta\gamma$ were readily reversible (Figure 5 – figure supplement 1), suggesting that most heterotrimers reassemble at the plasma membrane after activation ceases. In order to assess changes in heterotrimer trafficking to endosomes without interference from $G\beta\gamma$ translocation we incubated cells with agonist under conditions permissive for vesicular trafficking, then washed with antagonist to inactivate receptors and allow heterotrimers to reassemble prior to measuring BRET. Under these conditions no significant changes in bystander BRET were observed at any endosome compartment after 30 minutes of receptor stimulation (**Figure 5E** [↗](#)). Similar results were obtained when stimulation was maintained for 15 minutes or 5 minutes (Figure 5 – figure supplement 2), although small increases were apparent in a few instances. Taken together these results indicate that activation of receptors and G proteins does not substantially change heterotrimer abundance on the surface of endosomes.

Discussion

While the mechanisms involved in the biosynthesis, chaperoning and trafficking of nascent G protein heterotrimers are fairly well understood ([Gabay et al., 2011](#) [↗](#); [Marrari et al., 2007](#) [↗](#); [Wedegaertner, 2012](#) [↗](#)), the mechanisms that regulate the subcellular distribution of heterotrimers after delivery to the plasma membrane have not been studied as extensively. Here we show that constitutive and activity-dependent endocytosis of G proteins is inefficient. Avoidance of endocytosis is likely to be important for maintaining a high density of heterotrimers at the plasma membrane, where much important signaling takes place. On the other hand, this limits the abundance of G proteins on the surface of endosomes. At present we can only speculate regarding the mechanism that limits G protein density on endocytic vesicles. Many endocytosis mechanisms, including clathrin-mediated endocytosis, rely on bulky coat proteins and adapters to induce membrane curvature and recruit cargo ([Doherty & McMahon, 2009](#) [↗](#)). One possibility is that heterotrimeric G proteins are simply excluded from nascent endocytic vesicles by steric occlusion. While large extracellular domains are known to impede endocytosis of membrane proteins ([DeGroot et al., 2018](#) [↗](#)), a similar relationship has not been demonstrated for intracellular domains. Our finding that the small mNG-HRas ct probe is loaded onto endocytic vesicles more efficiently than mNG- β_1 -containing heterotrimers is consistent with this idea. It is noteworthy that the monomeric G proteins HRas and NRas are also less abundant on endosomes than the plasma membrane, and therefore are separated from internalized growth factor receptors ([Pinilla-Macua et al., 2016](#) [↗](#); [Surve et al., 2021](#) [↗](#)).

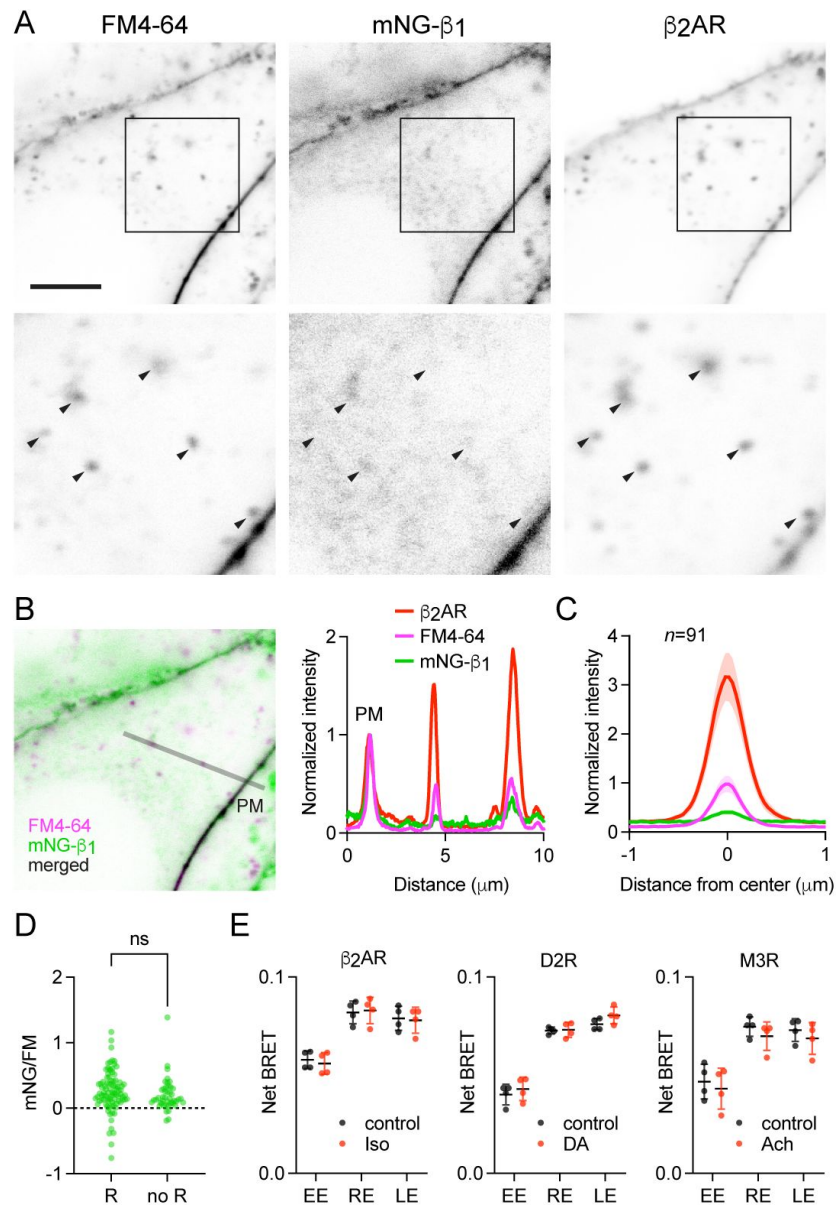


Figure 5 with 2 supplements

Receptor activation does not change G protein endocytosis.

(A) mNG-β₁ colocalizes with newly internalized endocytic vesicles labeled with FM4-64 and SNAP-tagged β₂ adrenergic receptor (β₂AR) labeled with Alexa Fluor 674; scale bar is 5 μm. Cells were stimulated with 10 μM isoproterenol for 15 minutes to induce β₂AR internalization. (B) A fluorescence intensity line profile for mNG-β₁, FM4-64 and β₂AR normalized to the peak value of each label at the plasma membrane (PM). (C) Mean mNG-β₁, FM4-64 and β₂AR fluorescence intensity line profiles drawn across vesicles, normalized to fluorescence intensity at the plasma membrane for each marker; mean ± 95% CI; $n=91$ vesicles/cells. (D) Normalized peak mNG-β₁/FM4-64 did not differ between vesicles that contained receptors (R; $n=91$) and vesicles formed by constitutive endocytosis (no R; $n=45$); n.s., not significant, $P=0.20$, unpaired t-test. (E) Bystander BRET between HiBit-β₁ and markers of early endosomes (EE), recycling endosomes (RE) and late endosomes (LE) was unchanged after 30 minutes of receptor activation with isoproterenol (Iso; 10 μM), dopamine (DA; 100 μM) or acetylcholine (Ach; 100 μM) compared to the DPBS vehicle alone (control); mean ± SD, $n=4$ independent experiments; no agonist-treated group was significantly different from the control, paired t-test with a false discovery rate (FDR) of 1%.

Some studies using overexpressed G protein subunits have suggested that a large pool of G proteins is located on intracellular membranes, including the Golgi apparatus (Chisari et al., 2007 [↗](#); Saini et al., 2007 [↗](#); Tsutsumi et al., 2009 [↗](#)), whereas others have indicated a distribution that is dominated by the plasma membrane (Crouthamel et al., 2008 [↗](#); Evanko et al., 2000 [↗](#); Marrari et al., 2007 [↗](#); Takida & Wedegaertner, 2003 [↗](#)). A likely factor contributing to these discrepant results is the stoichiometry of overexpressed subunits, as neither $G\alpha$ nor $G\beta\gamma$ traffic appropriately to the plasma membrane as free subunits (Wedegaertner, 2012 [↗](#)). Our gene-editing approach presumably maintains the native subunit stoichiometry, providing a more accurate representation of native G protein distribution. Our results show that endogenous G proteins are primarily located on the plasma membrane and are present on internal membranes at substantially lower levels. We identify the specific intracellular compartments where G proteins are found and show the relative abundance of G proteins on each compartment. Nascent heterotrimers are likely formed and lipid modified on the endoplasmic reticulum and Golgi apparatus (Wedegaertner, 2012 [↗](#)), yet few heterotrimers can be found on these compartments at any given moment, consistent with a relatively slow rate of turnover compared to forward trafficking during biosynthesis (Gabay et al., 2011 [↗](#)). Notably, when $G\beta\gamma$ dimers are expressed alone they accumulate on the endoplasmic reticulum (Michaelson et al., 2002 [↗](#); Takida & Wedegaertner, 2003 [↗](#)). That we detect almost no endogenous $G\beta\gamma$ on the endoplasmic reticulum supports our conclusion that the large majority of $G\beta\gamma$ in unstimulated HEK 293 cells is associated with $G\alpha$. Likewise, we found that few heterotrimers are associated with mitochondria, despite the fact that previous studies have demonstrated functional roles for G proteins on these organelles (Hewavitharana & Wedegaertner, 2012 [↗](#)). Our results suggest that GPCR signaling from intracellular compartments will generally have to be transduced by a lower density of G proteins. Consistent with these findings, a recent large-scale study assessing protein abundance on organelles in HEK 293 cells found that $G\beta_1$ is enriched on the plasma membrane and lysosomes but is not significantly enriched on endosomes, the Golgi apparatus, endoplasmic reticulum or mitochondria (Hein et al., 2023 [↗](#)).

Fully lipid-modified heterotrimers in their inactive state are unlikely to detach from membranes at a significant rate (Shahinian & Silvius, 1995 [↗](#)). Therefore, we infer from the presence of $G\beta_1$ on early, late, and recycling endosomes that heterotrimers undergo vesicle-mediated endocytosis in unstimulated cells and are not efficiently sorted to either the slow recycling pathway or the degradative pathway. Our imaging results also show that G proteins are apparently more abundant on late endosomes and lysosomes than on early endosomes, suggesting that they become concentrated as late endosomes mature. We do not know the fate of G proteins located on the surface of lysosomes. Since lysosomes may fuse with the plasma membrane under certain circumstances (Xu & Ren, 2015 [↗](#)), it is possible that this represents a route of G protein recycling to the plasma membrane. Our results are largely consistent with the hypothesis that G proteins passively follow bulk endocytic flow of membrane and suggest that at least some G proteins are recycled to the plasma membrane. We also cannot exclude the possibility that heterotrimers traffic between membrane compartments by mechanisms other than vesicular trafficking (Saini et al., 2009 [↗](#)). However, even if this is the case our conclusions that G proteins are internalized inefficiently and are present at lower density on most intracellular membranes are still valid.

Tagging endogenous proteins with HiBit and other luciferase fragments has proven to be useful for studying several aspects of GPCR signaling (White et al., 2020 [↗](#)). The cell lines we developed should prove useful for answering additional questions related to G protein regulation, such as possible non-vesicular translocation due to palmitate turnover (Saini et al., 2009 [↗](#); Wedegaertner & Bourne, 1994 [↗](#)), the role of ubiquitination (Dohlman & Campbell, 2019 [↗](#)), and localization in subcompartments not studied here. In addition, it may also be possible to use these cells in combination with targeted sensors to study endogenous G protein activation in different subcellular compartments. More broadly, our results show that subcellular localization of endogenous membrane proteins can be studied in living cells by adding a HiBit tag and performing bystander BRET mapping. Applied at large scale this approach would have some

advantages over fluorescent protein complementation (Cho et al., 2022 [↗](#)), most notably the ability to localize endogenous membrane proteins that are expressed at levels that are too low to permit fluorescence microscopy.

Our study is not without limitations. Our labeling strategy was designed to interfere as little as possible with heterotrimer function, but we cannot rule out the possibility that the tags we used to visualize and track G proteins had some influence on their trafficking. By labeling G β_1 subunits we cannot directly distinguish heterotrimers from free G $\beta\gamma$ dimers, complicating interpretation. This strategy also does not allow us to resolve heterotrimers containing different α subunits. It is quite possible that heterotrimers containing different α subunits could be subject to different trafficking mechanisms. Our conclusion that GPCR activation has no lasting effect on the subcellular distribution of G proteins rests on three representative receptors, chosen because they activate three of the four major G protein families. It is possible that other receptors will influence G protein distribution using mechanisms not shared by the receptors we studied. For example, a few receptors are thought to form relatively stable complexes with G $\beta\gamma$, which could provide a mechanism of trafficking to endosomes (Thomsen et al., 2016 [↗](#); Wehbi et al., 2013 [↗](#)). Finally, our study was limited to a single non-differentiated cell type. It would not be surprising to find that differentiated cells have mechanisms to regulate G protein trafficking and distribution that are not shared by the model cells we used (Calebiro et al., 2009 [↗](#); Kotowski et al., 2011 [↗](#); Lin et al., 2024 [↗](#); Puri et al., 2022 [↗](#)).

In summary, here we show that heterotrimeric G proteins are more abundant on the plasma membrane than on any intracellular compartment where they are thought to be important for signaling. Our results are likely to have functional implications for signaling from intracellular compartments. Receptor-G protein coupling is thought to be rate-limited by collision and G protein abundance (Hein et al., 2005 [↗](#)), and decreasing G protein expression is known to impair downstream signaling (Gabay et al., 2011 [↗](#); Schwindinger et al., 1997 [↗](#)). Signaling from endosomes and other compartments may thus be disadvantaged by a low density of G proteins. Further studies are warranted to examine the stoichiometry of receptors, G proteins, regulators, and effectors in different subcellular compartments and how this affects signaling.

Materials and Methods

Cell culture and transfection

Human embryonic kidney HEK 293 cells (ATCC; CRL-1573) were propagated in 100 mm dishes, on 6-well plates, or on 25 mm round coverslips in high glucose DMEM (Cytiva) and 10% fetal bovine serum (Cytiva) supplemented with penicillin streptomycin (Gibco). HEK 293T cells stably expressing mNG2(1-10) (Cho et al., 2022 [↗](#)) were kindly supplied by Manuel Leonetti (Chan Zuckerberg Biohub San Francisco). Cells were transfected in growth medium using linear polyethyleneimine MAX (Polysciences) at a nitrogen/phosphate ratio of 20 and were used for experiments 24-48 hours later. Up to 3.0 μ g of plasmid DNA was transfected in each well of a 6-well plate.

Gene editing

Ribonucleoprotein (RNP) complexes were assembled *in vitro* in IDT nuclease free duplex buffer from Alt-R™ crRNA, Alt-R™ tracrRNA and Alt-R™ S.p. Cas9 Nuclease V3 purchased from Integrated DNA Technologies (IDT). RNP and repair ssODNs (dissolved in nuclease-free water) were added to single-cell suspensions (10 μ l of 1.2×10^4 cells μ l⁻¹) and electroporated using a Neon™ Transfection device (Invitrogen) following the manufacturer's instructions. Cells were expanded and diluted into 48-well plates and grown for 3 weeks. Wells containing single cell colonies were duplicated into 12-well plates and screened for HiBit expression by mixing crude lysates with purified LgBit protein (Promega) and measuring luminescence in the presence of 5 μ M furimazine. After clone

expansion genomic DNA was extracted using the GeneJET Genomic DNA purification kit (ThermoFisher) and used as a template for amplicon sequencing. Sequencing primers were designed to span the editing site and to produce amplicons less than 500 base pairs in length. Amplicon sequencing was performed by Azenta Life Sciences (Amplicon-EZ) and analyzed using CRISPResso2. Cell lines used for experiments had correctly edited alleles but were hemizygous due to competing repair mechanisms. The human *GNB1* gene was targeted at a site corresponding to the N-terminus of the G β_1 protein; the sequence 5'-TGAGTGAGCTTGACCAGTTA-3' was incorporated into the crRNA, and the same gRNA was used to produce both HiBit- β_1 and mNG- β_1 cell lines. The ssODN homology-directed repair (HDR) template sequence for mNG- β_1 cells was: ATCTCACATTCTTGAAGGTGGCATTGAAGAGCACTAAGATCGGAAGATG**ACCGAGCTCAAC** **TTCAAGGAGTGGCAAAGGCCTTTACCGATATGATGGCGGAAGCGGTGTGTCCGGCTG** **GCGGCTGTTCAAGAAGATTCTGGCGGAAGCAGTGAGCTTGACCAGCTTAGACAGGAGGC** CGAGCAACTTAAGAACCAGA, with the mNG2(11) and HiBit tag sequences in **bold** font, and GGSG and GGS linkers in *italic* font. The repair template sequence for HiBit- β_1 cells was: TTTCAGATCTCACATTCTTGAAGGTGGCATTGAAGAGCACTAAGATCGGAAGAT**GGTGAGC** **GGCTGGCGGCTGTTCAAGAAGATTAGCGGCGGAAGCGGTAGTGAGCTTGACCAGCTTAG** **ACAGGAGGCCGAGCAACTTAAGAACCAGATTCGAG**, with the HiBit tag sequence in **bold** font, and GGSG linker in *italic* font. For both ssODNs a silent mutation (underlined sequence) was introduced to ablate the PAM site. HiBit was included in the repair template for producing mNG- β_1 cells to enable screening for edited clones using luminescence.

SDS-PAGE

Pelleted cells were mixed with Laemmli buffer (Bio-Rad), and proteins were separated on 4 to 15% SDS polyacrylamide gradient gels (Bio-Rad) then transferred to polyvinylidene difluoride (PVDF) membranes (Millipore Sigma). HiBit-tagged proteins were detected using the NanoGlo HiBit Blotting kit (Promega) following the manufacturer's instructions, and membranes were imaged using an Amersham Imager 600.

Plasmids

The following plasmids were used as received from Addgene: mRuby-Golgi-7 (GalT; #55865), mRuby2-Rab5a-7 (#55911), mCherry-Rab7a-7 (#55127), mCherry-Rab11a-7 (#55124), pmCherry-2xFYVE (#140050), mCherry-TGNP-N-10 (#55145), Lamp1-mScarlet-I(#98827). Venus-2xFYVE was made by replacing mCherry in pmCherry-2xFYVE with Venus using *NheI* and *BsrGI*. mRuby2-MOA was made by replacing Venus in Venus-MOA using *NheI* and *BglIII*. mRuby2-PTP1b was made by replacing Venus in Venus-PTP1b using *NheI* and *BsrGI*. CMV-LgBit was made by amplifying LgBit from pBiT1.1-N (Promega) and ligating into pcDNA3.1 (+) using *HindIII* and *XhoI*. SNAPf- β_2 AR, SNAPf-D2R, SNAPf-M3R and D2S-Nluc were kindly provided by Jonathan Javitch (Columbia University). The Nluc-EPAC-VV cyclic AMP sensor was kindly provided by Kirill Martemyanov (University of Florida). Venus-Kras, Venus-PTP1b, Venus-MOA, Venus-rab5a, Venus-rab7a, Venus-rab11a and memGRKct-Venus were described previously (Hollins et al., 2009 [\[1\]](#); Lan et al., 2012 [\[2\]](#)). All plasmids were verified by automated sequencing.

Imaging

Imaging was performed on a Leica SP8 laser scanning confocal microscope using a 63 $\times$ 1.40 NA oil immersion objective. Cells grown on 25 mm round coverslips were transferred to a steel imaging chamber and imaged in HEPES Imaging (HI) buffer which contained 150 mM NaCl, 10 mM NaHEPES, 5 mM glucose, 2.5 mM KCl, 1.2 mM CaCl₂, 1 mM MgCl₂ (pH 7.2). All imaging was carried out at room temperature with the exception of the experiment shown in Figure 4 – figure supplement 1, which was carried out at 37°C. For colocalization of mNG- β_1 and red organelle markers 0.2 μ g of each marker was transfected per coverslip; mNG- β_1 was excited at 488 nm and detected at 495-545 nm, and red markers were excited at 552 nm and detected at 565-665 nm. Lysosomes were stained with either LysoView 633 (Biotium; 1:1,000 in growth medium for 15

minutes at 37°C) or 10,000 m.w. CF 640 dextran (Biotium; 25 $\mu\text{g ml}^{-1}$ overnight at 37°C followed by a 60-minute chase); both dyes were excited at 638 nm and detected at 650–700 nm. For simultaneous imaging of mNG- β_1 , FM4-64 and CellMask Deep Red, cells were placed in HI buffer containing 1:1,000 CellMask Deep Red (Invitrogen) for 15 minutes at room temperature, then returned to culture medium containing 5 μM FM4-64 (a.k.a. SynaptoRed; Calbiochem) and incubated at 37°C for 15 minutes. Imaging was then performed in HI buffer containing 5 μM FM4-64. For simultaneous imaging of mNG- β_1 , FM4-64 and $\beta_2\text{AR}$, cells were transfected with 1 μg of SNAPf- $\beta_2\text{AR}$, stained in HI buffer containing 5 μM SNAP-Surface Alexa 647 (New England Biolabs) for 15 minutes at room temperature, then returned to culture medium containing 10 μM isoproterenol and 5 μM FM4-64 and incubated at 37°C for 15 minutes. Imaging was then performed in HI buffer containing 5 μM FM4-64. CellMask Deep Red and SNAP-Surface Alexa 647 were excited at 638 nm and detected at 650–700 nm; mNG- β_1 and FM4-64 were excited at 488 nm and detected at 500–570 nm and 675–755 nm, respectively.

Image analysis

Signal/background ratios for the plasma membrane and endosomes (**Figure 3C**) were calculated using mean fluorescence values from rectangular (for the plasma membrane) and round (for endosomes) regions of interest (ROIs) surrounding the structures and nearby cytosol (for background). A single plasma membrane ROI and 1–3 endosome ROIs were sampled per cell/image. Mean fluorescence intensity line profiles were extracted from 3 μm lines centered on vesicles as absolute fluorescence intensity (**Figure 3B**), or fluorescence intensity normalized to the mean intensity of a nearby section of plasma membrane (**Figure 4C** and **Figure 5C**). Vesicles that contained and did not contain internalized receptors were compared (**Figure 5D**) by dividing the peak mNG- β_1 signal by the peak FM4-64 signal for each vesicle; both signals were first normalized to their respective plasma membrane signals and subjected to background subtraction. A single vesicle was sampled per cell/image. All image analysis was carried out using ImageJ and raw images. For construction of figures images were exported as .TIF files with or without uniform contrast enhancement applied by ImageJ.

BRET

For bystander BRET mapping of HiBit- β_1 localization cells were transfected in 6-well plates with 1 μg per well of a Venus-tagged compartment marker and 0.1 μg per well of CMV-LgBit. For measurements cells were resuspended in Dulbecco's phosphate buffered saline (DPBS). For long-term agonist stimulation (**Figure 5E**) HiBit- β_1 cells expressing CMV-LgBit (0.1 μg per well) and either SNAPf- $\beta_2\text{AR}$, SNAPf-D2R or SNAPf-M3R (0.5 μg per well) were incubated with agonist for 30 minutes in the incubator, then washed and resuspended in DPBS containing antagonist prior to reading BRET. Agonists were isoproterenol (10 μM), dopamine (100 μM) and acetylcholine (100 μM); antagonists were ICI 118551 (10 μM), haloperidol (10 μM) and atropine (10 μM); all small molecule ligands were obtained from Millipore Sigma or Cayman Chemical. For functional validation of mNG- β_1 cells, D2R-Nluc (50 ng per well) was transfected, and cells were resuspended in permeabilization buffer (KPS) containing 140 mM KCl, 10 mM NaCl, 1 mM MgCl_2 , 0.1 mM Potassium EGTA, 20 mM NaHEPES (pH 7.2), 10 mg ml^{-1} high-purity digitonin and 2U ml^{-1} apyrase. Kinetic BRET measurements were made from permeabilized cells during sequential injection of dopamine (100 mM) and GDP (100 mM). For functional validation of HiBit- γ_1 cells, SNAPf- $\beta_2\text{AR}$, SNAPf-D2R or SNAPf-M3R (0.5 μg per well), CMV-LgBit (0.1 μg per well) and memGRK3ct-Venus (0.5 μg per well) were transfected, and cells were resuspended in DPBS. Kinetic BRET measurements were made from intact cells during sequential injection of agonists and antagonists at the concentrations listed above. All BRET measurements were made in buffer solutions containing the substrate furimazine (Promega or ChemShuttle; 1:1,000 from a 5 mM stock dissolved in 90% ethanol/10% glycerol). Steady-state BRET and luminescence measurements were made using a Mithras LB940 photon-counting plate reader (Berthold Technologies GmbH) running MicroWin2000 software. Kinetic BRET measurements were made using a Polarstar Optima plate reader (BMG Labtech) running BMG Optima version 2.20R2 software. Raw BRET signals were

calculated as the emission intensity at 520–545 nm divided by the emission intensity at 475–495 nm. Net BRET signals were calculated as the raw BRET signal minus the raw BRET signal measured from cells expressing only the donor.

Statistical analysis

All statistical testing was carried out using GraphPad Prism version 10.1.1. Comparison of mNG signals in vesicles with and without receptors (**Figure 5D** [↗](#)) was made using an unpaired t-test. Comparison of endosome bystander signals with and without agonist treatment (**Figure 5E** [↗](#)) was made using paired t-tests with a false discovery rate (FDR) of 1% (method of Benjamini, Krieger and Yekutieli).

Data availability

All study data are included in the article and source data files. Cell lines and plasmids generated for this study are freely available upon request from the corresponding author. No unique code or software was used for the study.

Acknowledgements

We thank Manuel Leonetti for providing HEK 293 cells expressing mNG2(1-10) and Kirill Martemyanov and Jonathan Javitch for providing plasmid DNA. This study was supported by NIH grant GM145284 (N.A.L.).

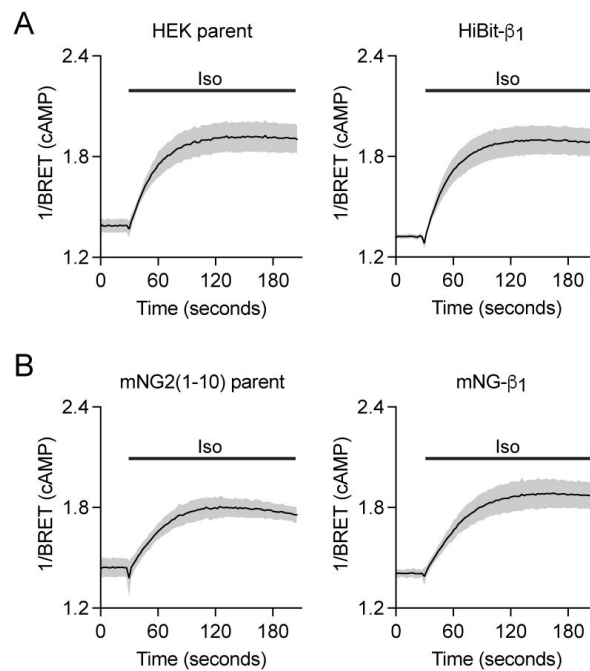


Figure 1 – figure supplement 1

Receptor-mediated accumulation of cyclic AMP (cAMP) is similar in HiBit- β_1 , mNG- β_1 and parent cell lines.

Activation of endogenous β adrenergic receptors with isoproterenol (Iso; 10 μ M) produces similar increases in cAMP as indicated by the Nluc-EPAC-VV cAMP sensor in **(A)** parental HEK and HiBit- β_1 cells and **(B)** parental mNG2(1-10) and mNG- β_1 cells; mean $\pm$ 95% CI of 20 replicates from $n=5$ independent experiments.

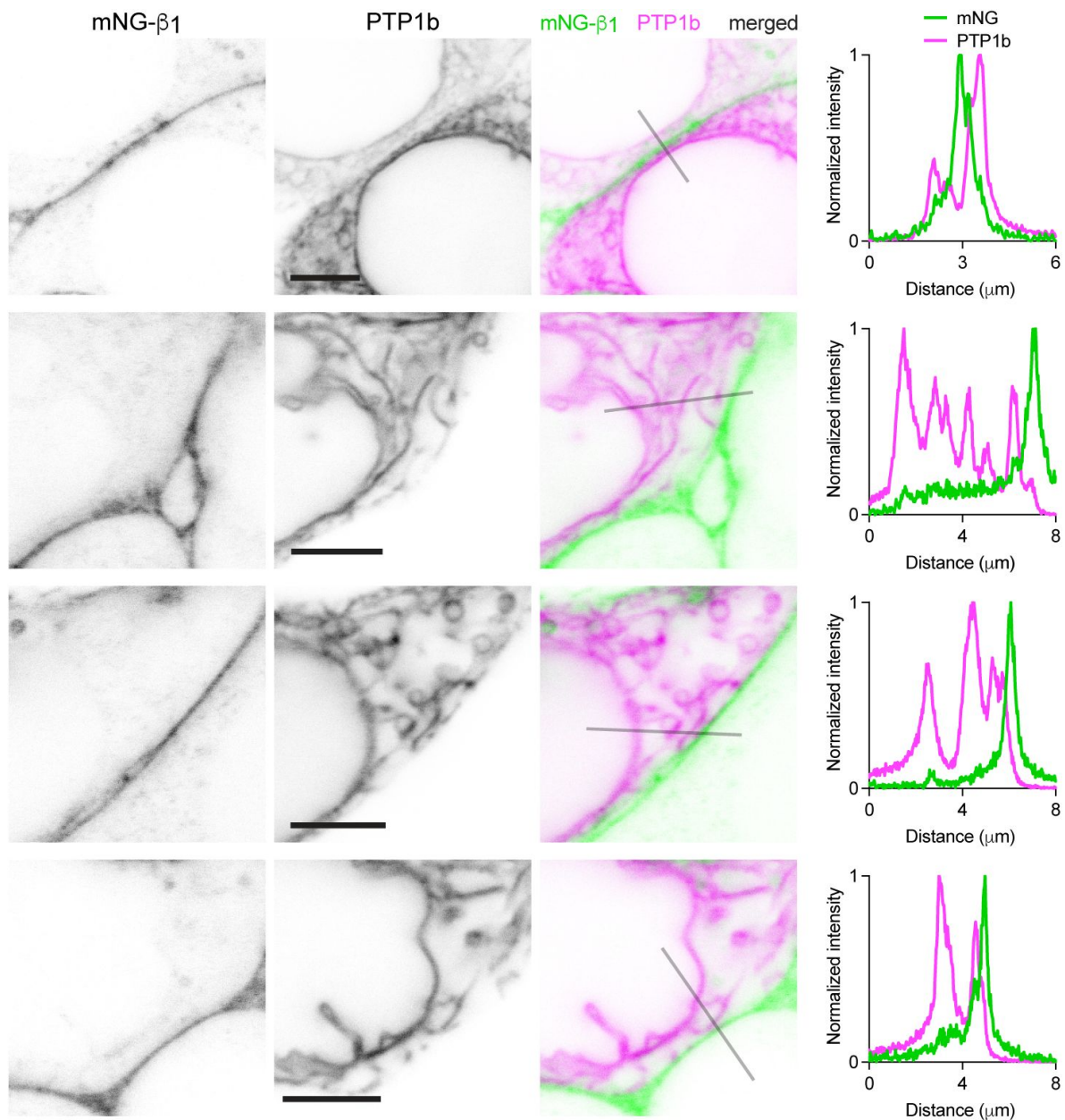


Figure 2 – figure supplement 1

G proteins are not abundant on the endoplasmic reticulum (ER).

Exemplanary images of mNG-β₁ cells coexpressing the ER marker mRuby2-PTP1b. Intensity line profiles depict fluorescence intensity normalized to the maximum value in each channel; scale bars are 5 μm.

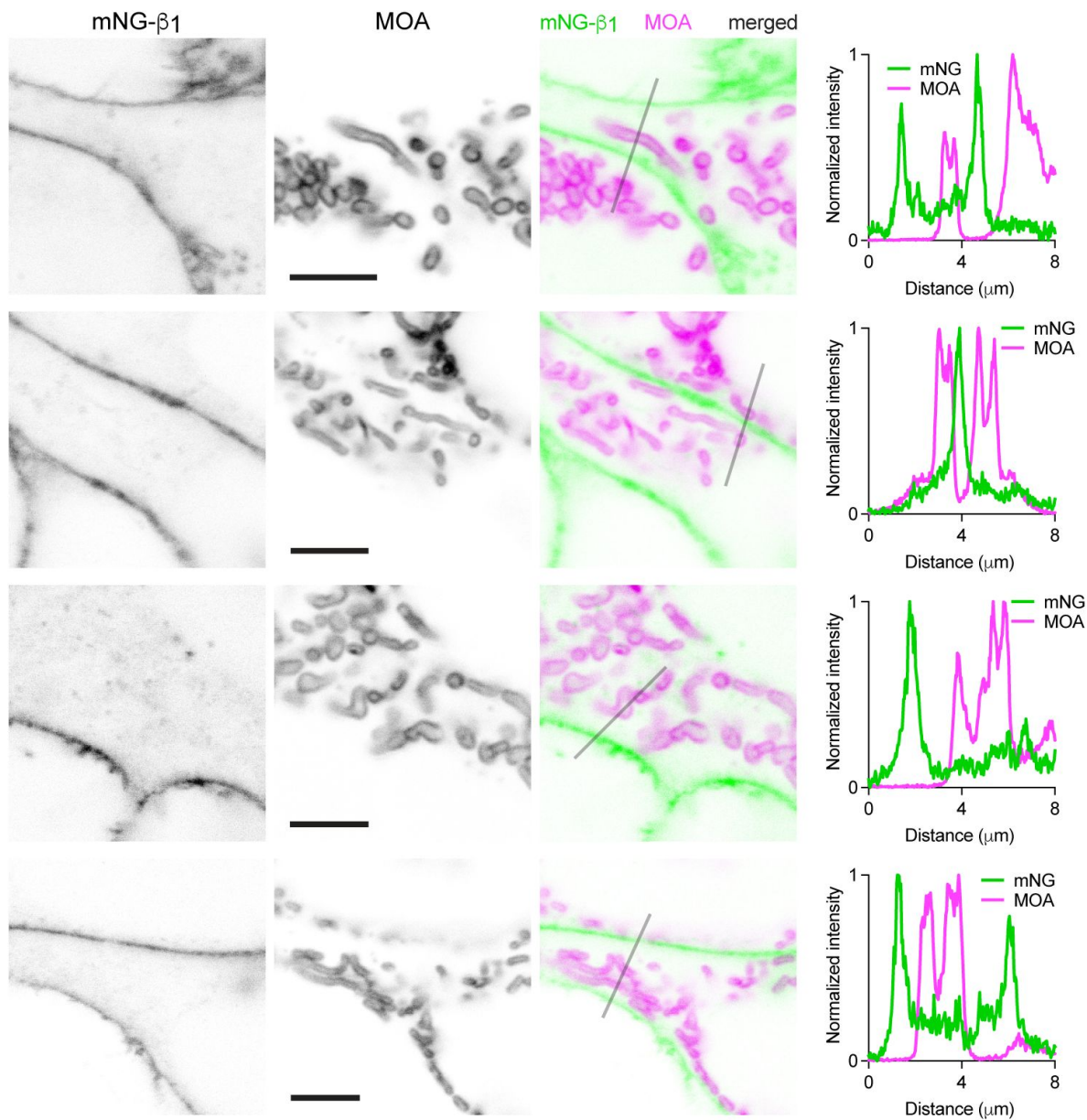


Figure 2 – figure supplement 2

G proteins are not abundant on mitochondria (MT).

Exemplary images of mNG- γ_1 cells coexpressing the MT marker mRuby2-MOA. Intensity line profiles depict fluorescence intensity normalized to the maximum value in each channel; scale bars are 5 μ m.

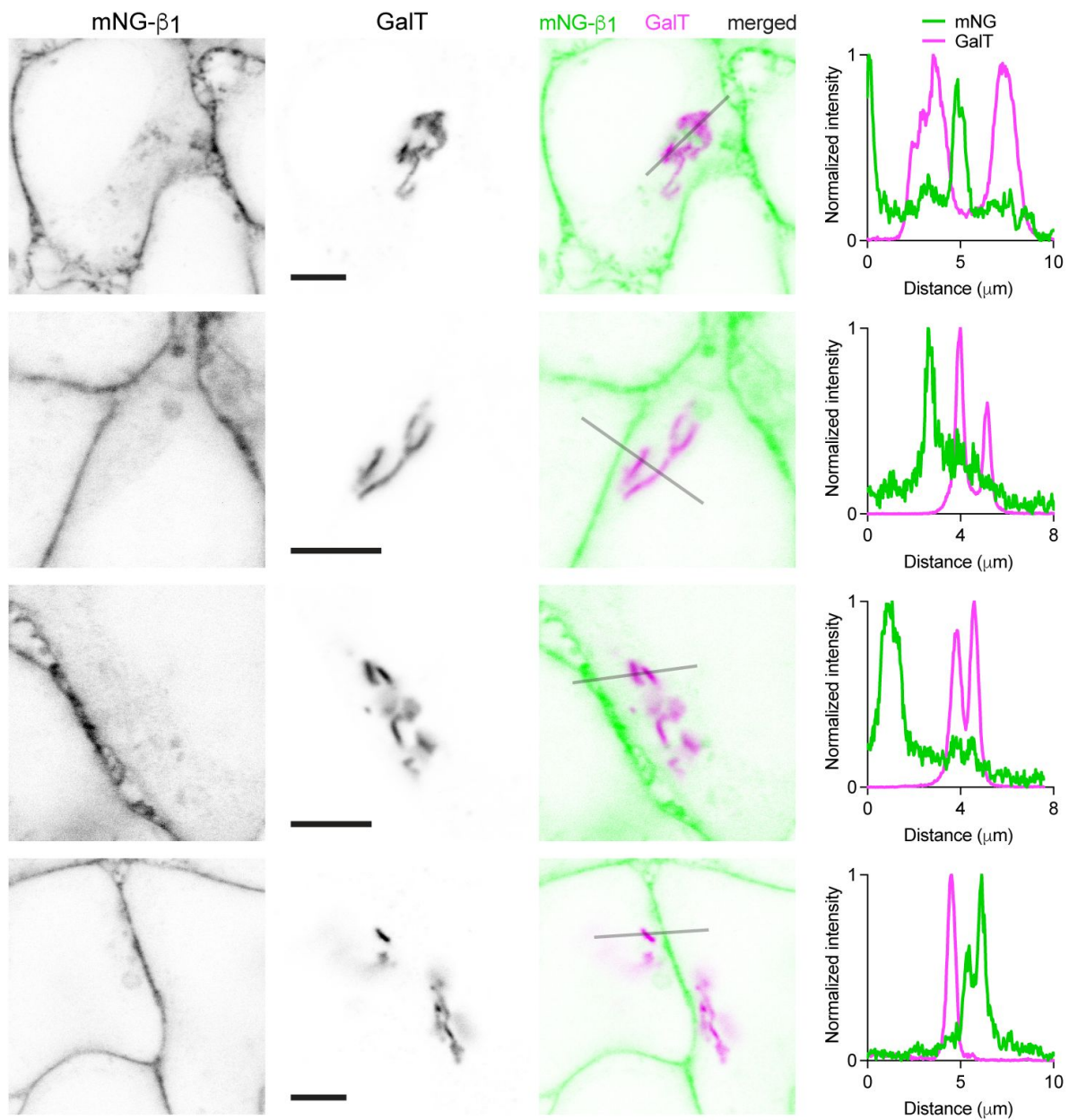


Figure 2 – figure supplement 3

G proteins are not abundant on the medial Golgi apparatus (GA).

Exemphary images of mNG-β₁ cells coexpressing the GA marker mRuby2-Golgi-7 (GalT). Intensity line profiles depict fluorescence intensity normalized to the maximum value in each channel; scale bars are 5 μm.

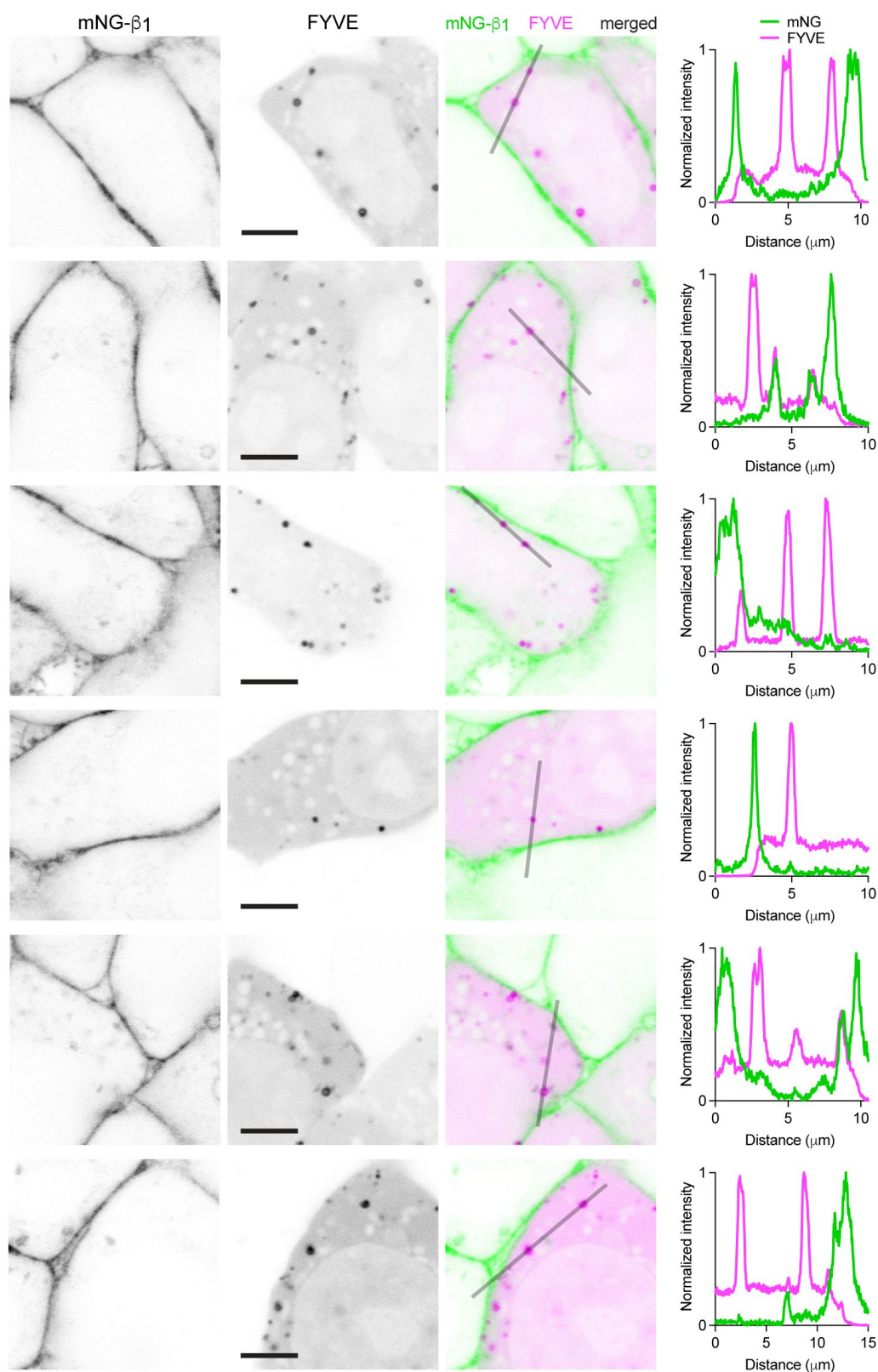


Figure 3 – figure supplement 1

G proteins colocalize with the early endosome marker FYVE on some endosomes.

Exemplary images of mNG-β₁ cells coexpressing the early endosome marker pmCherry-2xFYVE (FYVE). Intensity line profiles depict fluorescence intensity normalized to the maximum value in each channel; scale bars are 5 μm.

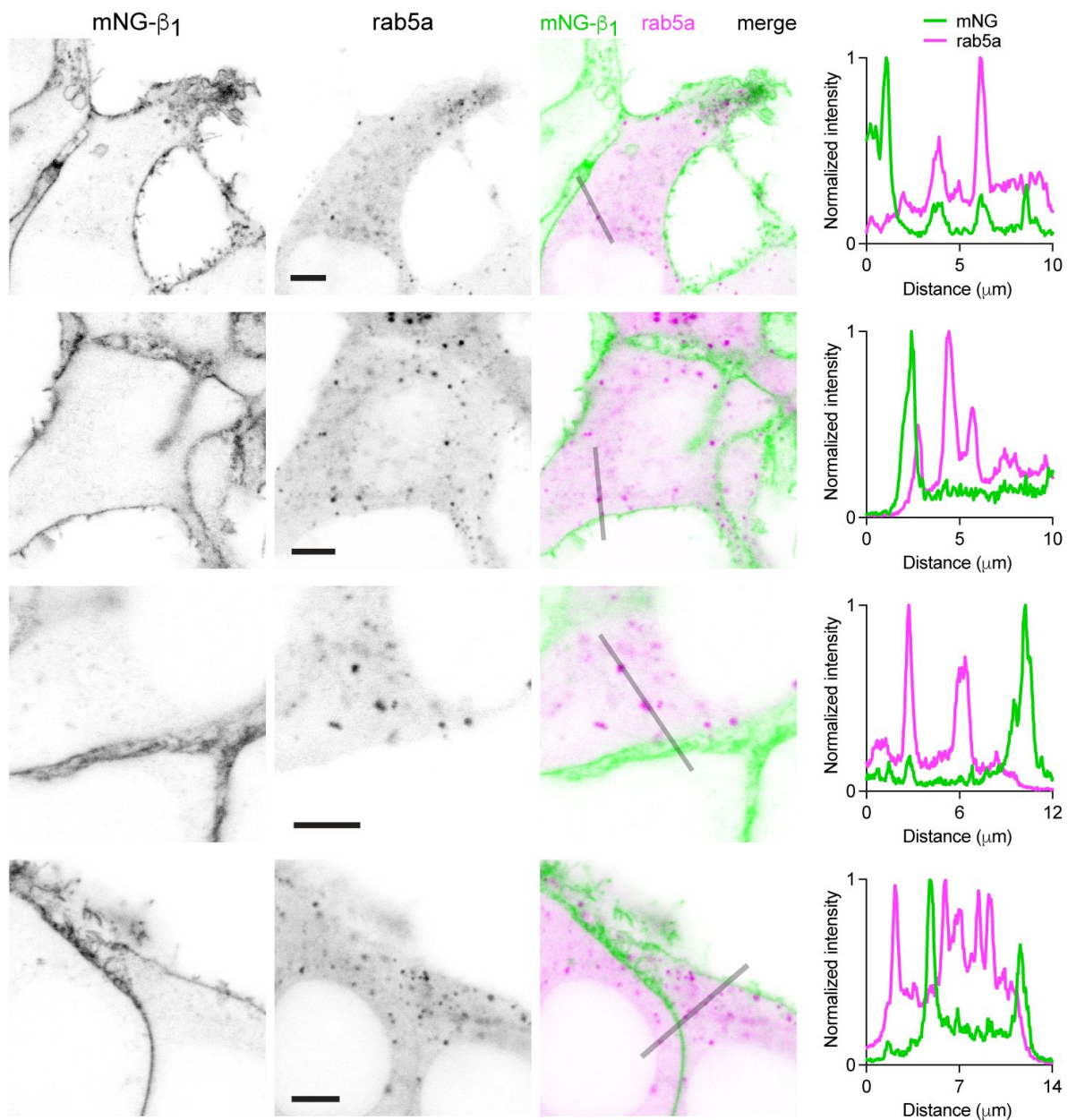


Figure 3 – figure supplement 2

G proteins colocalize with the early endosome marker rab5a on some endosomes.

Exemplary images of mNG-β₁ cells coexpressing the early endosome marker mRuby2-rab5a (rab5a). Intensity line profiles depict fluorescence intensity normalized to the maximum value in each channel; scale bars are 5 μm.

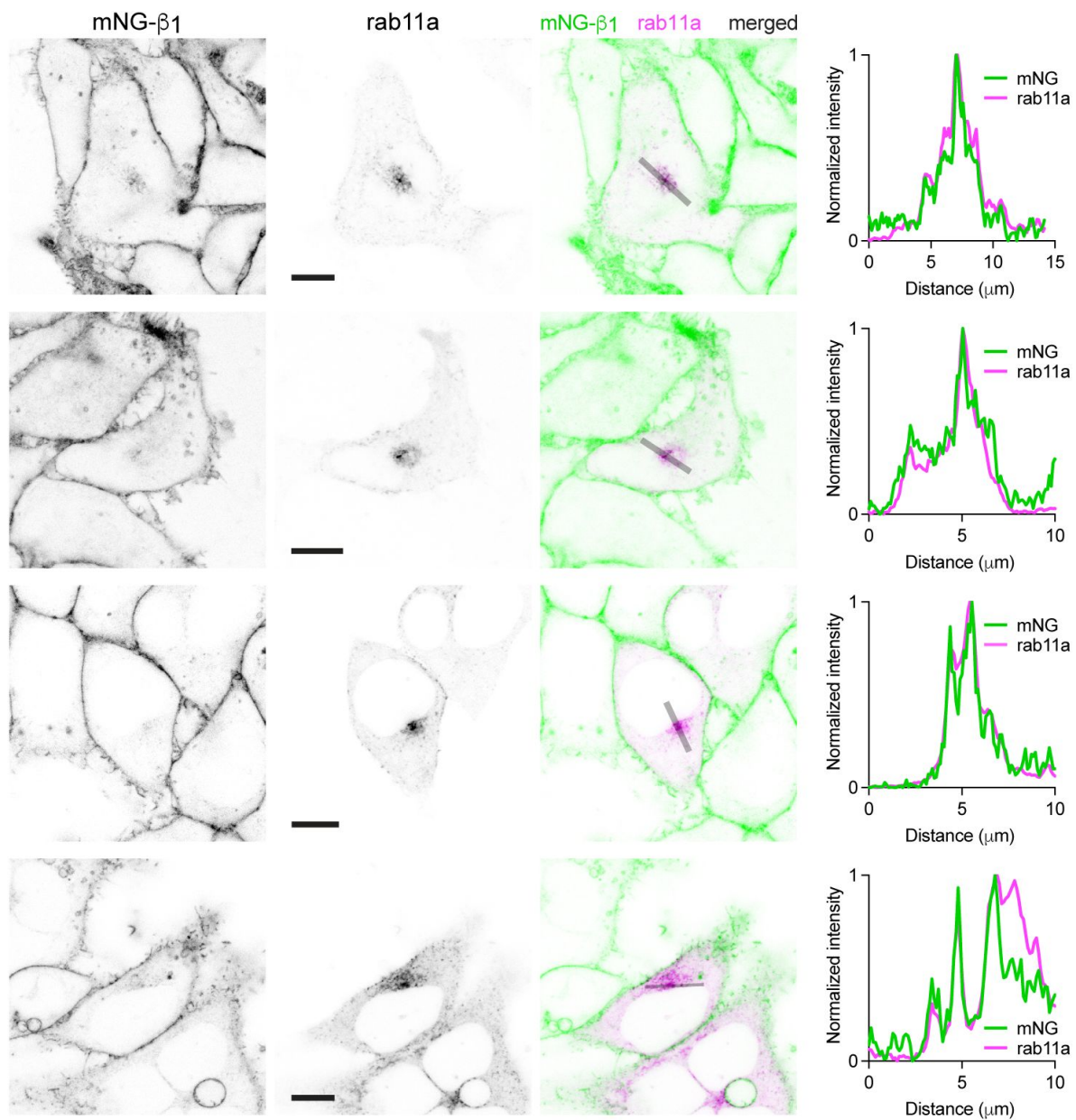


Figure 3 - figure supplement 3

G proteins colocalize with the recycling endosome marker rab11a in a perinuclear region.

Exemplary images of mNG- β_1 cells coexpressing the recycling endosome marker mCherry-rab11a (rab11a). Intensity line profiles depict fluorescence intensity normalized to the maximum value in each channel; scale bars are 10 μm .

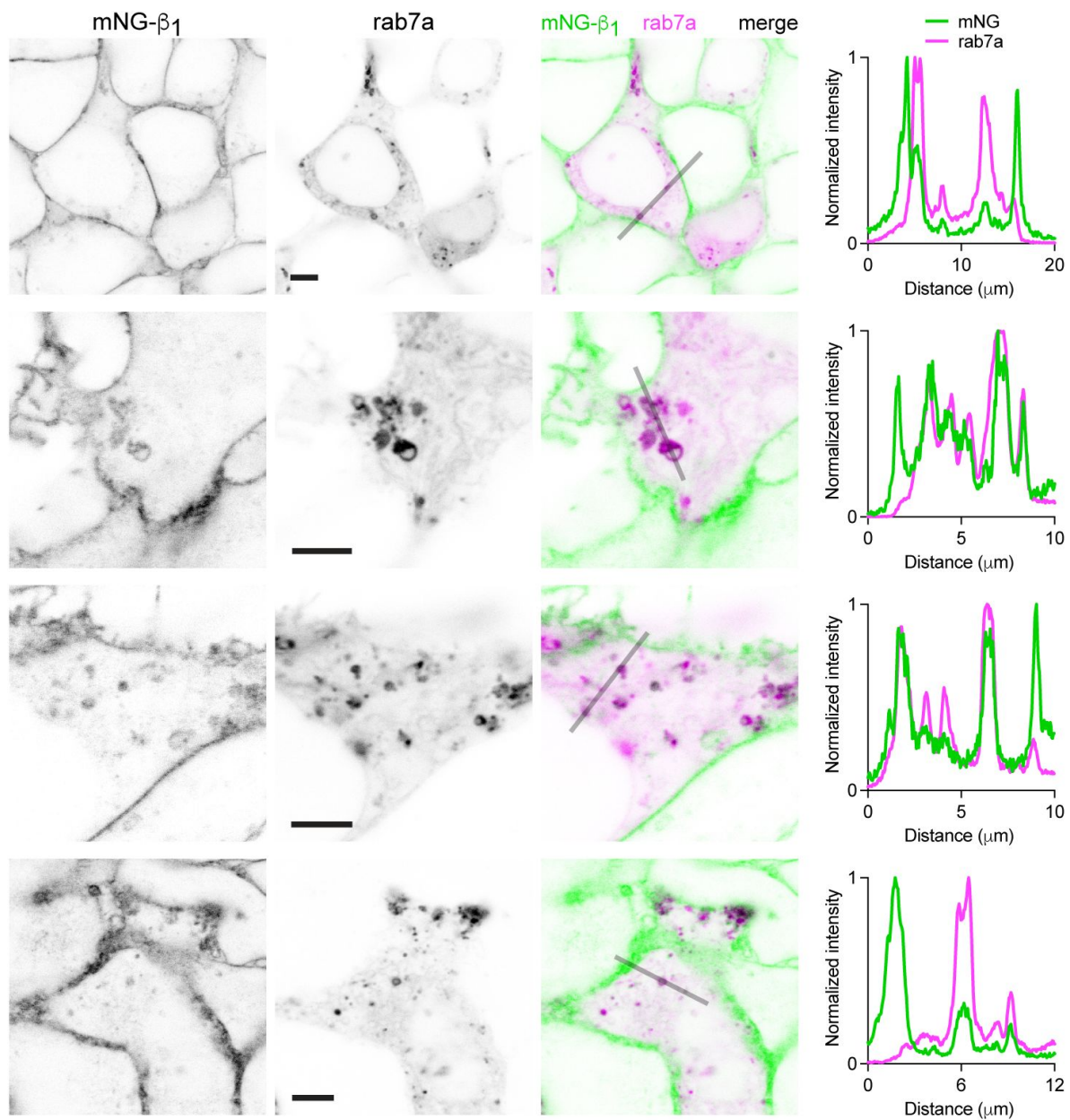


Figure 3 – figure supplement 4

G proteins colocalize with the late endosome marker rab7a on many endosomes.

Exemplary images of mNG-β₁ cells coexpressing the late endosome marker mCherry-rab7a (rab7a). Intensity line profiles depict fluorescence intensity normalized to the maximum value in each channel; scale bars are 5 μm.

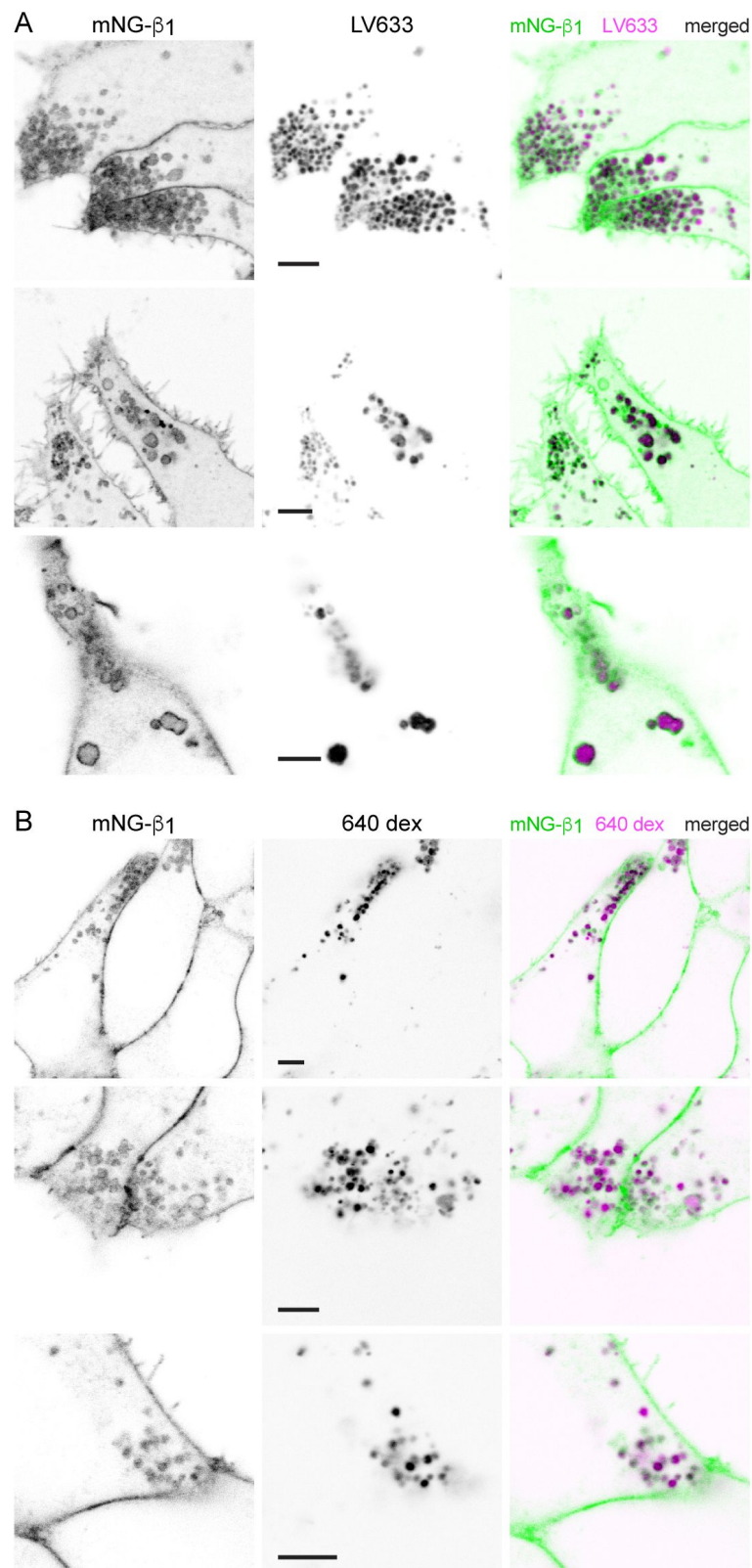


Figure 3 – figure supplement 5

G proteins are abundant on lysosomes.

(A) Exemplary images of mNG-β₁ cells stained with LysoView 633 (LV633); scale bars are 5 μm. (B) Exemplary images of mNG-β₁ cells incubated overnight with 10,000 m.w. CF 640 dextran (640 dex); scale bars are 5 μm.

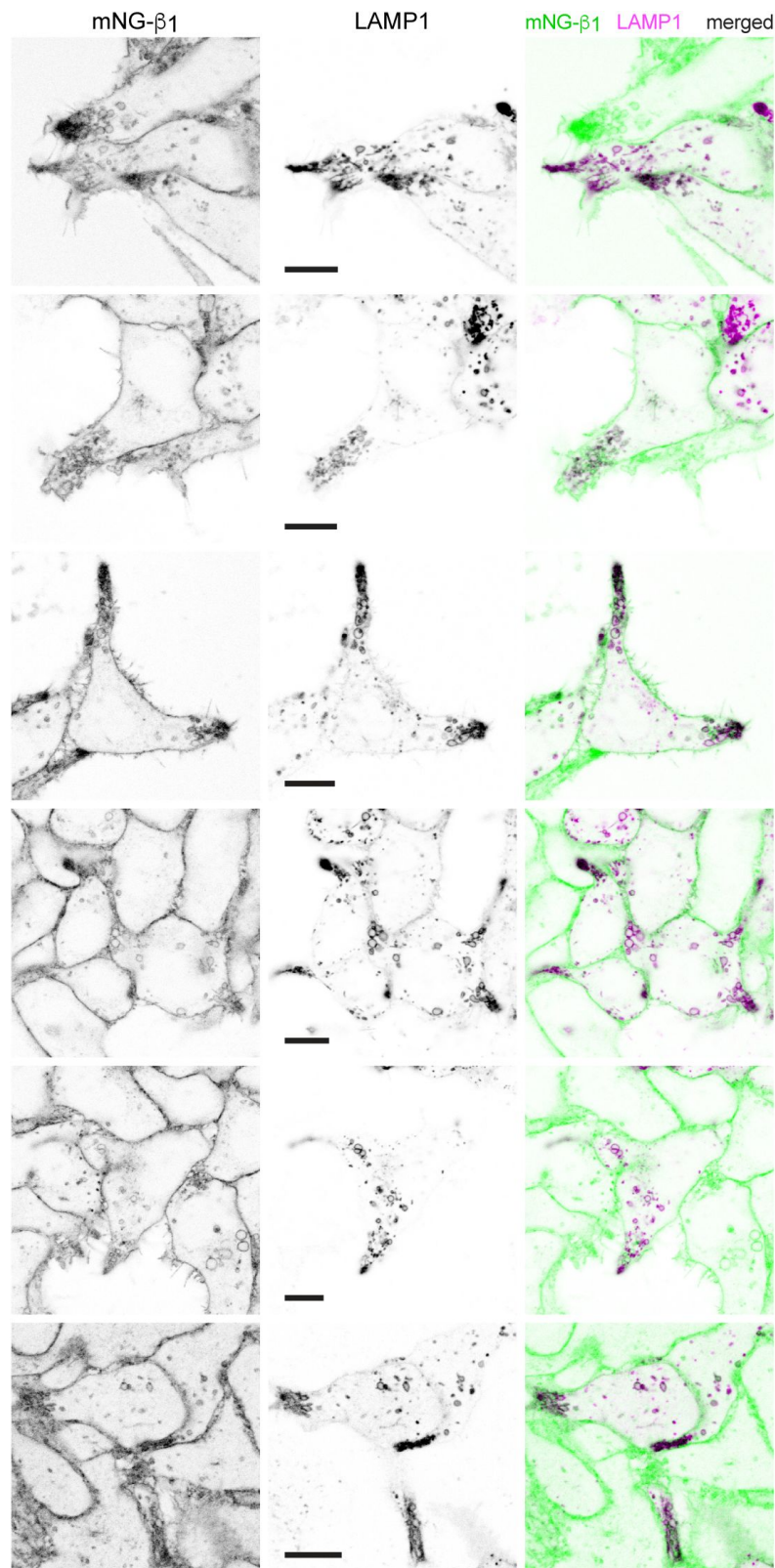


Figure 3 – figure supplement 6

G proteins colocalize with the lysosome marker LAMP1.

Exemplary images of mNG- γ_1 cells coexpressing the lysosome marker LAMP1-mScarlet; scale bars are 5 μ m.

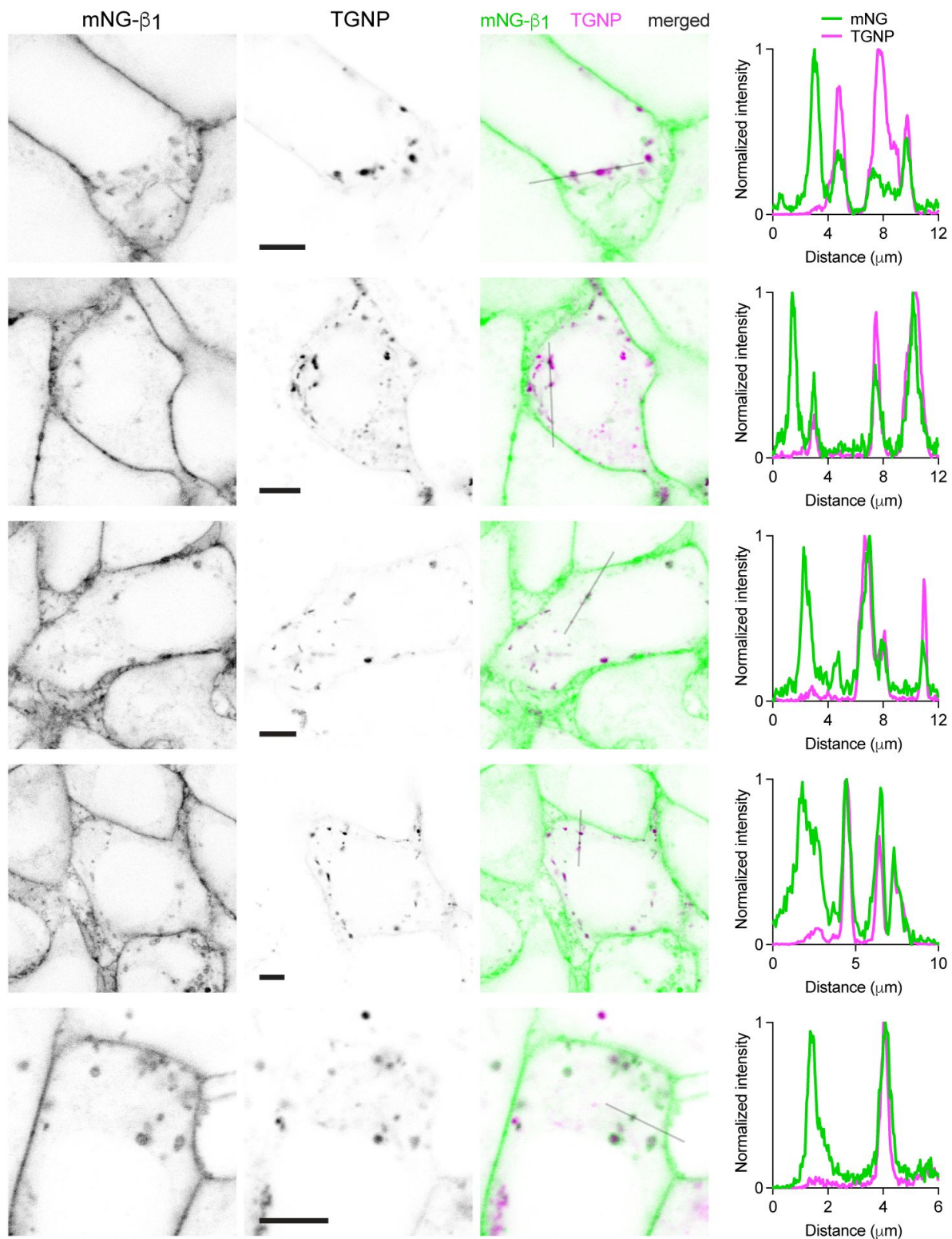


Figure 3 – figure supplement 7

G proteins colocalize with the trans-Golgi marker TGNP.

Exemplary images of mNG- γ_1 cells coexpressing the trans-Golgi marker mCherry-TGNP. Intensity line profiles depict fluorescence intensity normalized to the maximum value in each channel; scale bars are 5 μ m.

Figure 4 – figure supplement 2

mNG-HRas ct is less abundant on endocytic vesicles than the plasma membrane.

(A) mNG-HRas ct colocalizes with newly internalized endocytic vesicles labeled with FM4-64 (arrowheads); scale bar is 2 μ m. (B) A fluorescence intensity line profile for mNG-HRas ct and FM4-64 normalized to the peak value of each label at the plasma membrane (PM). (C) Mean mNG-HRas ct and FM4-64 fluorescence intensity line profiles drawn across vesicles, normalized to fluorescence intensity at the plasma membrane for each label; mean $\pm$ 95% CI; $n=78$ vesicles/cells.

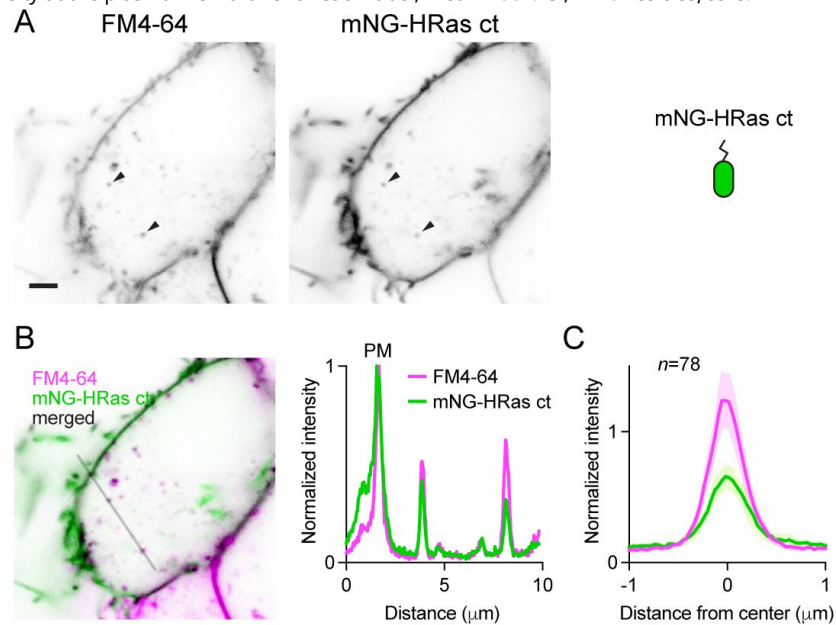
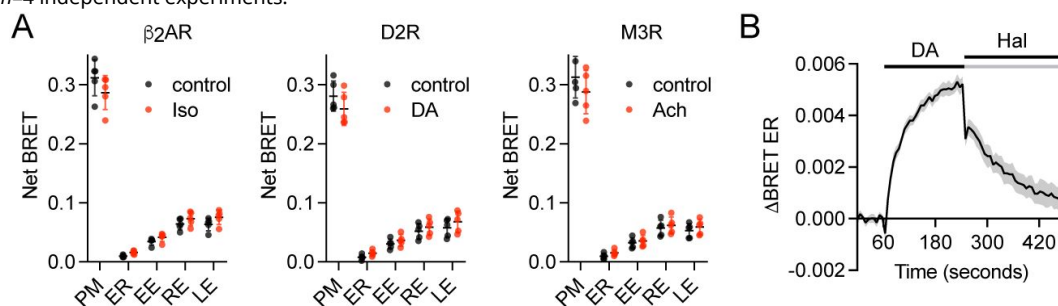


Figure 5 – figure supplement 1

Transient translocation of endogenous HiBit- β_1 from the plasma membrane to intracellular compartments *during* activation.

(A) Bystander BRET between HiBit- β_1 and a Venus-tagged marker of the plasma membrane (PM) decreased, whereas BRET between HiBit- β_1 and a Venus-tagged markers of the endoplasmic reticulum (ER), early endosomes (EE), recycling endosomes (RE) and late endosomes (LE) increased *during* receptor activation with isoproterenol (Iso; 10 μ M), dopamine (DA; 100 μ M) or acetylcholine (ACh; 100 μ M) compared to the DPBS vehicle alone (control); mean $\pm$ SD, $n=5$ independent experiments; all agonist-treated groups were significantly different from the control, paired t-test with a false discovery rate (FDR) of 1%. (B) Time course of bystander BRET between HiBit- β_1 and a Venus-tagged marker of the ER during activation of D2R with DA (100 μ M), followed by block of D2R with the antagonist haloperidol (Hal; 10 μ M); mean $\pm$ 95% CI of 24 replicates from $n=4$ independent experiments.



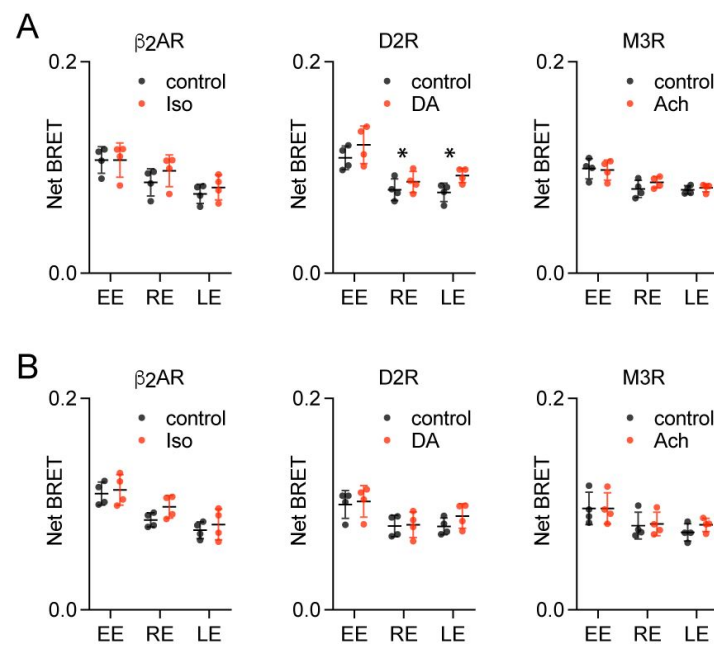


Figure 5 – figure supplement 2

G protein abundance on endosomes after GPCR and G protein activation.

Bystander BRET between HiBit- β_1 and markers of early endosomes (EE), recycling endosomes (RE) and late endosomes (LE) was largely unchanged after 5 minutes (**A**) or 15 minutes (**B**) of receptor activation with isoproterenol (Iso; 10 μ M), dopamine (DA; 100 μ M) or acetylcholine (Ach; 100 μ M) compared to the DPBS vehicle alone (control); mean $\pm$ SD, $n=4$ independent experiments; only two agonist-treated groups (indicated with *) were significantly different from the control, paired t-test with a false discovery rate (FDR) of 1%.

References

1. Akgoz M., Kalyanaraman V., Gautam N (2004) **Receptor-mediated reversible translocation of the G protein betagamma complex from the plasma membrane to the Golgi complex** *J Biol Chem* **279**:51541–51544 <https://doi.org/10.1074/jbc.M410639200>
2. Barak L. S., Warabi K., Feng X., Caron M. G., Kwatra M. M (1999) **Real-time visualization of the cellular redistribution of G protein-coupled receptor kinase 2 and beta-arrestin 2 during homologous desensitization of the substance P receptor** *J Biol Chem* **274**:7565–7569 <https://doi.org/10.1074/jbc.274.11.7565>
3. Benovic J. L., Bouvier M., Caron M. G., Lefkowitz R. J (1988) **Regulation of adenylyl cyclase-coupled beta-adrenergic receptors** *Annu Rev Cell Biol* **4**:405–428 <https://doi.org/10.1146/annurev.cb.04.110188.002201>
4. Betz W. J., Mao F., Smith C. B (1996) **Imaging exocytosis and endocytosis** *Curr Opin Neurobiol* **6**:365–371 [https://doi.org/10.1016/s0959-4388\(96\)80121-8](https://doi.org/10.1016/s0959-4388(96)80121-8)
5. Bowman S. L., Shiwarski D. J., Puthenveedu M. A (2016) **Distinct G protein-coupled receptor recycling pathways allow spatial control of downstream G protein signaling** *J Cell Biol* **214**:797–806 <https://doi.org/10.1083/jcb.201512068>
6. Calebiro D., Nikolaev V. O., Gagliani M. C., de Filippis T., Dees C., Tacchetti C., Persani L., Lohse M. J. (2009) **Persistent cAMP-signals triggered by internalized G-protein-coupled receptors** *PLoS biology* **7**:e1000172–e1000172 <https://doi.org/10.1371/journal.pbio.1000172>
7. Calebiro D., Nikolaev V. O., Persani L., Lohse M. J (2010) **Signaling by internalized G-protein-coupled receptors** *Trends Pharmacol Sci* **31**:221–228 <https://doi.org/10.1016/j.tips.2010.02.002>
8. Chisari M., Saini D. K., Kalyanaraman V., Gautam N (2007) **Shuttling of G protein subunits between the plasma membrane and intracellular membranes** *The Journal of biological chemistry* **282**:24092–24098 <https://doi.org/10.1074/jbc.M704246200>
9. Cho N. H. *et al.* (2022) **OpenCell: Endogenous tagging for the cartography of human cellular organization** *Science* **375** <https://doi.org/10.1126/science.abi6983>
10. Crouthamel M., Thiyagarajan M. M., Evanko D. S., Wedegaertner P. B (2008) **N-terminal polybasic motifs are required for plasma membrane localization of Galpha(s) and Galpha(q)** *Cellular signalling* **20**:1900–1910 <https://doi.org/10.1016/j.cellsig.2008.06.019>
11. DeGroot A. C. M., Busch D. J., Hayden C. C., Mihelic S. A., Alpar A. T., Behar M., Stachowiak J. C (2018) **Entropic Control of Receptor Recycling Using Engineered Ligands** *Biophys J* **114**:1377–1388 <https://doi.org/10.1016/j.bpj.2018.01.036>
12. Doherty G. J., McMahon H. T (2009) **Mechanisms of endocytosis** *Annu Rev Biochem* **78**:857–902 <https://doi.org/10.1146/annurev.biochem.78.081307.110540>
13. Dohlman H. G., Campbell S. L (2019) **Regulation of large and small G proteins by ubiquitination** *J Biol Chem* **294**:18613–18623 <https://doi.org/10.1074/jbc.REV119.011068>

14. Eichel K., von Zastrow M. (2018) **Subcellular Organization of GPCR Signaling** *Trends Pharmacol Sci* **39**:200–208 <https://doi.org/10.1016/j.tips.2017.11.009>
15. Evanko D. S., Thiyagarajan M. M., Wedegaertner P. B (2000) **Interaction with Gbetagamma is required for membrane targeting and palmitoylation of Galpha(s) and Galpha(q)** *The Journal of biological chemistry* **275**:1327–1336
16. Feng S., Sekine S., Pessino V., Li H., Leonetti M. D., Huang B (2017) **Improved split fluorescent proteins for endogenous protein labeling** *Nat Commun* **8** <https://doi.org/10.1038/s41467-017-00494-8>
17. Ferrandon S., Feinstein T. N., Castro M., Wang B., Bouley R., Potts J. T., Gardella T. J., Vilardaga J. P (2009) **Sustained cyclic AMP production by parathyroid hormone receptor endocytosis** *Nat Chem Biol* **5**:734–742 <https://doi.org/10.1038/nchembio.206>
18. Gabay M., Pinter M. E., Wright F. A., Chan P., Murphy A. J., Valenzuela D. M., Yancopoulos G. D., Tall G. G (2011) **Ric-8 proteins are molecular chaperones that direct nascent G protein α subunit membrane association** *Sci Signal* **4** <https://doi.org/10.1126/scisignal.2002223>
19. Gilman A. G (1987) **G proteins: transducers of receptor-generated signals** *Annual review of biochemistry* **56**:615–649 <https://doi.org/10.1146/annurev.bi.56.070187.003151>
20. Hammond G. R., Balla T (2015) **Polyphosphoinositide binding domains: Key to inositol lipid biology** *Biochim Biophys Acta* **1851**:746–758 <https://doi.org/10.1016/j.bbalip.2015.02.013>
21. Hanyaloglu A. C., von Zastrow M. (2008) **Regulation of GPCRs by endocytic membrane trafficking and its potential implications** *Annu Rev Pharmacol Toxicol* **48**:537–568
22. Hein M. Y. *et al.* (2023) **Global organelle profiling reveals subcellular localization and remodeling at proteome scale** *bioRxiv* <https://doi.org/10.1101/2023.12.18.572249>
23. Hein P., Frank M., Hoffmann C., Lohse M. J., Bunemann M (2005) **Dynamics of receptor/G protein coupling in living cells** *Embo J* **24**:4106–4114
24. Hewavitharana T., Wedegaertner P. B (2012) **Non-canonical signaling and localizations of heterotrimeric G proteins** *Cellular signalling* **24**:25–34 <https://doi.org/10.1016/j.cellsig.2011.08.014>
25. Hillenbrand M., Schori C., Schöppe J., Plückthun A (2015) **Comprehensive analysis of heterotrimeric G-protein complex diversity and their interactions with GPCRs in solution** *Proc Natl Acad Sci U S A* **112**:E1181–1190 <https://doi.org/10.1073/pnas.1417573112>
26. Hollins B., Kuravi S., Digby G. J., Lambert N. A (2009) **The c-terminus of GRK3 indicates rapid dissociation of G protein heterotrimers** *Cellular signalling* **21**
27. Hynes T. R., Mervine S. M., Yost E. A., Sabo J. L., Berlot C. H (2004) **Live cell imaging of Gs and the beta2-adrenergic receptor demonstrates that both alphas and beta1gamma7 internalize upon stimulation and exhibit similar trafficking patterns that differ from that of the beta2-adrenergic receptor** *The Journal of biological chemistry* **279**:44101–44112 <https://doi.org/10.1074/jbc.M405151200>
28. Irannejad R., Pessino V., Mika D., Huang B., Wedegaertner P. B., Conti M., von Zastrow M. (2017) **Functional selectivity of GPCR-directed drug action through location bias** *Nature chemical biology* **13**:799–806 <https://doi.org/10.1038/nchembio.2389>

29. Irannejad R. *et al.* (2013) **Conformational biosensors reveal GPCR signalling from endosomes** *Nature* **495**:534–538 <https://doi.org/10.1038/nature12000>
30. Kotowski S. J., Hopf F. W., Seif T., Bonci A., von Zastrow M. (2011) **Endocytosis promotes rapid dopaminergic signaling** *Neuron* **71**:278–290 <https://doi.org/10.1016/j.neuron.2011.05.036>
31. Krumins A. M., Gilman A. G (2006) **Targeted knockdown of G protein subunits selectively prevents receptor-mediated modulation of effectors and reveals complex changes in non-targeted signaling proteins** *J Biol Chem* **281**:10250–10262 <https://doi.org/10.1074/jbc.M511551200>
32. Lan T.-H., Liu Q., Li C., Wu G., Lambert N. A (2012) **Sensitive and high resolution localization and tracking of membrane proteins in live cells with BRET**. *Traffic (Copenhagen Denmark)* **13**:1450–1456 <https://doi.org/10.1111/j.1600-0854.2012.01401.x>
33. Lazar A. M., Irannejad R., Baldwin T. A., Sundaram A. B., Gutkind J. S., Inoue A., Dessauer C. W., Von Zastrow M. (2020) **G protein-regulated endocytic trafficking of adenylyl cyclase type 9** *eLife* **9** <https://doi.org/10.7554/eLife.58039>
34. Lin T. Y., Mai Q. N., Zhang H., Wilson E., Chien H. C., Yee S. W., Giacomini K. M., Olgin J. E., Irannejad R (2024) **Cardiac contraction and relaxation are regulated by distinct subcellular cAMP pools** *Nat Chem Biol* **20**:62–73 <https://doi.org/10.1038/s41589-023-01381-8>
35. Marrari Y., Crouthamel M., Irannejad R., Wedegaertner P. B (2007) **Assembly and Trafficking of Heterotrimeric G Proteins †** *Biochemistry* **46**:7665–7677 <https://doi.org/10.1021/bi700338m>
36. Martin B. R., Lambert N. A (2016) **Activated G Protein Gα subs/sub Samples Multiple Endomembrane Compartments** *Journal of Biological Chemistry* **291**:20295–20302 <https://doi.org/10.1074/jbc.M116.729731>
37. Michaelson D., Ahearn I., Bergo M., Young S., Philips M (2002) **Membrane trafficking of heterotrimeric G proteins via the endoplasmic reticulum and Golgi** *Mol Biol Cell* **13**:3294–3302 <https://doi.org/10.1091/mbc.e02-02-0095>
38. Mullershausen F., Zecri F., Cetin C., Billich A., Guerini D., Seuwen K (2009) **Persistent signaling induced by FTY720-phosphate is mediated by internalized S1P1 receptors** *Nat Chem Biol* **5**:428–434 <https://doi.org/10.1038/nchembio.173>
39. Pierce K. L., Premont R. T., Lefkowitz R. J (2002) **Seven-transmembrane receptors** *Nat Rev Mol Cell Biol* **3**:639–650
40. Pinilla-Macua I., Watkins S. C., Sorkin A (2016) **Endocytosis separates EGF receptors from endogenous fluorescently labeled HRas and diminishes receptor signaling to MAP kinases in endosomes** *Proc Natl Acad Sci U S A* **113**:2122–2127 <https://doi.org/10.1073/pnas.1520301113>
41. Puri N. M., Romano G. R., Lin T. Y., Mai Q. N., Irannejad R (2022) **The organic cation transporter 2 regulates dopamine D1 receptor signaling at the Golgi apparatus** *eLife* **11** <https://doi.org/10.7554/eLife.75468>
42. Ransnäs L. A., Svoboda P., Jasper J. R., Insel P. A (1989) **Stimulation of beta-adrenergic receptors of S49 lymphoma cells redistributes the alpha subunit of the stimulatory G protein between cytosol and membranes** *Proceedings of the National Academy of Sciences of the United States of America* **86**:7900–7903

43. Saini D. K., Chisari M., Gautam N (2009) **Shuttling and translocation of heterotrimeric G proteins and Ras** *Trends in pharmacological sciences* **30**:278–286 <https://doi.org/10.1016/j.tips.2009.04.001>
44. Saini D. K., Kalyanaraman V., Chisari M., Gautam N (2007) **A family of G protein $\beta\gamma$ subunits translocate reversibly from the plasma membrane to endomembranes on receptor activation** *J Biol Chem* **282**:24099–24108 <https://doi.org/10.1074/jbc.M701191200>
45. Scarselli M., Donaldson J. G (2009) **Constitutive internalization of G protein-coupled receptors and G proteins via clathrin-independent endocytosis** *The Journal of biological chemistry* **284**:3577–3585 <https://doi.org/10.1074/jbc.M806819200>
46. Schwindinger W. F., Reese K. J., Lawler A. M., Gearhart J. D., Levine M. A (1997) **Targeted disruption of Gnas in embryonic stem cells** *Endocrinology* **138**:4058–4063 <https://doi.org/10.1210/endo.138.10.5439>
47. Schwinn M. K., Machleidt T., Zimmerman K., Eggers C. T., Dixon A. S., Hurst R., Hall M. P., Encell L. P., Binkowski B. F., Wood K. V (2018) **CRISPR-Mediated Tagging of Endogenous Proteins with a Luminescent Peptide** *ACS Chemical Biology* **13**:467–474 <https://doi.org/10.1021/acscchembio.7b00549>
48. Shahinian S., Silvius J. R (1995) **Doubly-lipid-modified protein sequence motifs exhibit long-lived anchorage to lipid bilayer membranes** *Biochemistry* **34**:3813–3822 <https://doi.org/10.1021/bi00011a039>
49. Slepak V. Z., Hurley J. B (2008) **Mechanism of light-induced translocation of arrestin and transducin in photoreceptors: interaction-restricted diffusion** *IUBMB life* **60**:2–9 <https://doi.org/10.1002/iub.7>
50. Stenmark H (2009) **Rab GTPases as coordinators of vesicle traffic** *Nat Rev Mol Cell Biol* **10**:513–525 <https://doi.org/10.1038/nrm2728>
51. Surve S., Watkins S. C., Sorkin A (2021) **EGFR-RAS-MAPK signaling is confined to the plasma membrane and associated endorecycling protrusions** *J Cell Biol* **220** <https://doi.org/10.1083/jcb.202107103>
52. Takida S., Wedegaertner P. B (2003) **Heterotrimer Formation, Together with Isoprenylation, Is Required for Plasma Membrane Targeting of G $\beta\gamma$** *Journal of Biological Chemistry* **278**:17284–17290 <https://doi.org/10.1074/jbc.M213239200>
53. Thomsen A. R. B. *et al.* (2016) **GPCR-G Protein- β -Arrestin Super-Complex Mediates Sustained G Protein Signaling** *Cell* **166**:907–919 <https://doi.org/10.1016/j.cell.2016.07.004>
54. Tsutsumi R., Fukata Y., Noritake J., Iwanaga T., Perez F., Fukata M (2009) **Identification of G protein α subunit-palmitoylating enzyme** *Mol Cell Biol* **29**:435–447 <https://doi.org/10.1128/mcb.01144-08>
55. Tsvetanova N. G., Irannejad R., von Zastrow M. (2015) **GPCR signaling via heterotrimeric G proteins from endosomes** *Journal of Biological Chemistry* **290**:jbc.R114.617951–jbc.R617114.617951 <https://doi.org/10.1074/jbc.R114.617951>
56. von Zastrow M., Kobilka B. K. (1992) **Ligand-regulated internalization and recycling of human beta 2-adrenergic receptors between the plasma membrane and endosomes containing transferrin receptors** *J Biol Chem* **267**:3530–3538

57. Wedegaertner P. B (2012) **G protein trafficking** *Subcellular Biochemistry* **63**:193–223 https://doi.org/10.1007/978-94-007-4765-4_11
58. Wedegaertner P. B., Bourne H. R (1994) **Activation and depalmitoylation of Gs alpha** *Cell* **77**:1063–1070
59. Wedegaertner P. B., Bourne H. R., von Zastrow M. (1996) **Activation-induced subcellular redistribution of Gs alpha** *Molecular biology of the cell* **7**:1225–1233
60. Wehbi V. L., Stevenson H. P., Feinstein T. N., Calero G., Romero G., Vilardaga J. P (2013) **Noncanonical GPCR signaling arising from a PTH receptor-arrestin-Gβγ complex** *Proc Natl Acad Sci U S A* **110**:1530–1535 <https://doi.org/10.1073/pnas.1205756110>
61. White C. W., Caspar B., Vanyai H. K., Pflieger K. D. G., Hill S. J (2020) **CRISPR-Mediated Protein Tagging with Nanoluciferase to Investigate Native Chemokine Receptor Function and Conformational Changes** *Cell Chem Biol* **27**:499–510 <https://doi.org/10.1016/j.chembiol.2020.01.010>
62. Xu H., Ren D (2015) **Lysosomal physiology** *Annu Rev Physiol* **77**:57–80 <https://doi.org/10.1146/annurev-physiol-021014-071649>

Editors

Reviewing Editor

Felix Campelo

Institute of Photonic Sciences, Barcelona, Spain

Senior Editor

Felix Campelo

Institute of Photonic Sciences, Barcelona, Spain

Reviewer #1 (Public review):

Summary:

The manuscript by Jang et al. describes the application of new methods to measure the localization GTP-binding signaling proteins (G proteins) on different membrane structures in a model mammalian cell line (HEK293). G proteins mediate signaling by receptors found at the cell surface (GPCRs), with evidence from the last 15 years suggesting that GPCRs can induce G-protein mediated signaling from different membrane structures within the cell, with variation in signal localization leading to different cellular outcomes. While it has been clearly shown that different GPCRs efficiently traffic to various intracellular compartments, it is less clear whether G proteins traffic in the same manor, and whether GPCR trafficking facilitates "passenger" G protein trafficking. This question was a blind spot in the burgeoning field of GPCR localized signaling in need of careful study, and the results obtained will serve as an important guide post for further work in this field.

The extent to which G proteins localize to different membranes within the cell is the main experimental question tested in this manuscript. This question is pursued by through two distinct methods, both relying on genetic modification of the G-beta subunit with a tag. In one method, G-beta is modified with a small fragment of the fluorescent protein mNG, which combines with the larger mNG fragment to form a fully functional fluorescent protein to facilitate protein trafficking by fluorescent microscopy. This approach was combined with

expression of fluorescent proteins directed to various intracellular compartments (different types of endosomes, lysosome, endoplasmic reticulum, golgi, mitochondria) to look for colocalization of G-beta with these markers. These experiments showed compelling evidence that G-beta co-localizes with markers at the plasma membrane and the lysosome, with weak or absent co-localization for other markers. A second method for measuring localization relied on fusing G-beta with a small fragment from a miniature luciferase (HiBit) that combines with a larger luciferase fragment (LgBit) to form an active luciferase enzyme. Localization of G-beta (and luciferase signal) was measured using a method known as bystander BRET, which relies on expression of a fluorescent protein BRET acceptor in different cellular compartments. Results using bystander BRET supported findings from fluorescence microscopy experiments. These methods for tracking G protein localization were also used to probe other questions. The activation of GPCRs from different classes had virtually no impact on the localization of G-beta, suggesting that GPCR activation does not result in shuttling of G proteins through the endosomal pathway with activated receptors.

In the revised version of this manuscript the authors have performed informative and important new experiments in addition to adding new text to address conceptual questions. These new data and discussions are commendable and address most or all of the weaknesses listed in the initial review.

Strengths:

The question probed in this study is quite important and, in my opinion, understudied by the pharmacology community. The results presented here are an important call to be cognizant of the localization of GPCR coupling partners in different cellular compartments. Abundant reports of endosomal GPCR signaling need to consider how the impact of lower G protein abundance on endosomal membranes will affect the signaling responses under study.

*The work presented is carefully executed, with seemingly high levels of technical rigor. These studies benefit from probing the experimental questions at hand using two different methods of measurement (fluorescent microscopy and bystander BRET). The observation that both methods arrive at the same (or a very similar) answer inspires confidence about the validity of these findings.

Weaknesses:

*As noted by the authors, they do not demonstrate that the tagged G-beta is predominantly found within heterotrimeric G protein complexes. In the revised manuscript the authors have added new discussion text on why it is likely that G-beta is mostly found in complexes. This line of reasoning is convincing, although more robust experimental methods for assessing the assembly status of G-beta could be a valuable target for future experimental developments.

<https://doi.org/10.7554/eLife.97033.2.sa3>

Reviewer #2 (Public review):

This study assess the subcellular distribution of a major G protein subunit (Gβ1) when expressed at an endogenous level in a well-studied model cell system (293 cells). The approach elegantly extends a gene editing strategy described by Leonetti's group and combines it with a FRET-based proximity assay to detect the presence of endogenously tagged Gβ1 on membrane compartments of 293 cells. The authors achieve their goal, and the data are convincing and interesting. The authors do a nice job of integrating their results with previous work in the field. The methods are now sufficiently well-described to enable other investigators to apply or adapt them in future studies.

Reviewer #3 (Public review):**Summary:**

This article addresses an important and interesting question concerning intracellular localization and dynamics of endogenous G proteins. The fate and trafficking of G protein-coupled receptors (GPCRs) have been extensively studied but so far little is known about the trafficking routes of their partner G proteins that are known to dissociate from their respective receptors upon activation of the signaling pathway. Authors utilize modern cell biology tools including genome editing and bystander bioluminescence resonance energy transfer (BRET) to probe intracellular localization of G proteins in various membrane compartments in steady state and also upon receptor activation. Data presented in this manuscript shows that while G proteins are mostly present on the plasma membrane, they can be also detected in endosomal compartments, especially in late endosomes and lysosomes. This distribution, according to data presented in this study, seems not to be affected by receptor activation. These findings will have implications in further studies addressing GPCR signaling mechanisms from intracellular compartments.

Strengths:

The methods used in this study are adequate for the question asked. Especially use of genome-edited cells (for addition of the tag on one of the G proteins) is a great choice to prevent effects of overexpression. Moreover, use of bystander BRET allowed authors to probe intracellular localization of G proteins in a very high-throughput fashion. By combining imaging and BRET authors convincingly show that G proteins are very low abundant on early endosomes (also ER, mitochondria, and medial Golgi), however seem to accumulate on membranes of late endosomal compartments. Moreover, authors also looked at the dynamics of G protein trafficking by tracking them over multiple time points in different compartments.

Weaknesses:

While authors provide a novel dataset, many questions regarding G protein trafficking remain open. For example, it is not entirely clear which pathway is utilized to traffic G proteins from the plasma membrane to intracellular compartments. Additionally, future studies should also include more quantitative details considering G-protein distribution in different compartments as well as more detailed dynamic data on G protein internalization as well as intracellular trafficking kinetics.

<https://doi.org/10.7554/eLife.97033.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:**Reviewer #1 (Public Review):****Summary:**

The manuscript by Jang et al. describes the application of new methods to measure the localization of GTP-binding signaling proteins (G proteins) on different membrane

structures in a model mammalian cell line (HEK293). G proteins mediate signaling by receptors found at the cell surface (GPCRs), with evidence from the last 15 years suggesting that GPCRs can induce G-protein mediated signaling from different membrane structures within the cell, with variation in signal localization leading to different cellular outcomes. While it has been clearly shown that different GPCRs efficiently traffic to various intracellular compartments, it is less clear whether G proteins traffic in the same manner, and whether GPCR trafficking facilitates "passenger" G protein trafficking. This question was a blind spot in the burgeoning field of GPCR localized signaling in need of careful study, and the results obtained will serve as an important guidepost for further work in this field. The extent to which G proteins localize to different membranes within the cell is the main experimental question tested in this manuscript. This question is pursued through two distinct methods, both relying on genetic modification of the G-beta subunit with a tag. In one method, G-beta is modified with a small fragment of the fluorescent protein mNG, which combines with the larger mNG fragment to form a fully functional fluorescent protein to facilitate protein trafficking by fluorescent microscopy. This approach was combined with the expression of fluorescent proteins directed to various intracellular compartments (different types of endosomes, lysosome, endoplasmic reticulum, Golgi, mitochondria) to look for colocalization of G-beta with these markers. These experiments showed compelling evidence that G-beta co-localizes with markers at the plasma membrane and the lysosome, with weak or absent co-localization for other markers. A second method for measuring localization relied on fusing G-beta with a small fragment from a miniature luciferase (HiBit) that combines with a larger luciferase fragment (LgBit) to form an active luciferase enzyme. Localization of Gbeta (and luciferase signal) was measured using a method known as bystander BRET, which relies on the expression of a fluorescent protein BRET acceptor in different cellular compartments. Results using bystander BRET supported findings from fluorescence microscopy experiments. These methods for tracking G protein localization were also used to probe other questions. The activation of GPCRs from different classes had virtually no impact on the localization of G-beta, suggesting that GPCR activation does not result in the shuttling of G proteins through the endosomal pathway with activated receptors.

Strengths:

The question probed in this study is quite important and, in my opinion, understudied by the pharmacology community. The results presented here are an important call to be cognizant of the localization of GPCR coupling partners in different cellular compartments. Abundant reports of endosomal GPCR signaling need to consider how the impact of lower G protein abundance on endosomal membranes will affect the signaling responses under study.

The work presented is carefully executed, with seemingly high levels of technical rigor. These studies benefit from probing the experimental questions at hand using two different methods of measurement (fluorescent microscopy and bystander BRET). The observation that both methods arrive at the same (or a very similar) answer inspires confidence about the validity of these findings.

Weaknesses:

The rationale for fusing G-beta with either mNG2(11) or SmBit could benefit from some expansion. I understand the speculation that using the smallest tag possible may have the smallest impact on protein performance and localization, but plenty of researchers have fused proteins with whole fluorescent proteins to provide conclusions that have been confirmed by other methods. Many studies even use G proteins fused with fluorescent proteins or luciferases. Is there an important advantage to tagging G-beta with small tags? Is there evidence that G proteins with full-size protein tags behave

aberrantly? If the studies presented here would not have been possible without these CRISPR-based tagging approaches, it would be helpful to provide more context to make this clearer. Perhaps one factor would be interference from newly synthesized G proteins-fluorescent protein fusions en route to the plasma membrane (in the ER and Golgi).

There are several advantages to using small peptide tags that we did not fully explain. From a practical standpoint the most important advantage of using the HiBit tag instead of full-length Nanoluc is that it allows us to restrict luminescence output to cells transiently transfected with LgBit. In this way untransfected cells contribute no background signal. Although we did not take advantage of it here, this also applies to fluorescent protein complementation, and will be useful for visualizing proteins in individual cells within tissues. The HiBit tag also allows PAGE analysis by probing membranes with LgBit (as in Fig. 1). We are not aware of evidence that tagging Gb or Gg subunits on the N terminus results in aberrant behavior, while there is some evidence that Ga subunits tagged with full-size protein tags (in some positions) have altered functional properties (PMID: 16371464). We do think that editing endogenous genes is critical, as studies using transient overexpression (usually driven by strong promoters) have sometimes reported accumulation of tagged G proteins in the biosynthetic pathway (e.g., PMID: 17576765), as the reviewer suggests. Ga and Gbg appear to be mutually dependent on each other for appropriate trafficking to the plasma membrane (reviewed in PMID: 23161140), therefore the native (presumably matched) stoichiometry is likely to be critical.

To clarify this context the revised manuscript includes the following:

“For bioluminescence experiments we added the HiBit tag (Schwinn et al., 2018) and isolated clonal “HiBit-b1” cell lines. An advantage of this approach over adding a full-length Nanoluc luciferase is that it requires coexpression of LgBit to produce a complemented luciferase. This limits luminescence to cotransfected cells and thus eliminates background from untransfected cells.”

“Some studies using overexpressed G protein subunits have suggested that a large pool of G proteins is located on intracellular membranes, including the Golgi apparatus (Chisari et al., 2007; Saini et al., 2007; Tsutsumi et al., 2009), whereas others have indicated a distribution that is dominated by the plasma membrane (Crouthamel et al., 2008; Evanko, Thiyagarajan, & Wedegaertner, 2000; Marrari et al., 2007; Takida & Wedegaertner, 2003). A likely factor contributing to these discrepant results is the stoichiometry of overexpressed subunits, as neither Ga nor Gbg traffic appropriately to the plasma membrane as free subunits (Wedegaertner, 2012). Our gene-editing approach presumably maintains the native subunit stoichiometry, providing a more accurate representation of native G protein distribution.”

As noted by the authors, they do not demonstrate that the tagged G-beta is predominantly found within heterotrimeric G protein complexes. If there is substantial free G-beta, then many of the conclusions need to be reconsidered. Perhaps a comparison of immunoprecipitated tagged G beta vs immunoprecipitated supernatant, with blotting for other G protein subunits would be informative.

We do think that HiBit-b1 exists predominantly within heterotrimeric complexes, for several reasons. First, overexpression studies have shown that Gbg requires association with Ga to traffic to the plasma membrane, and that by itself Gbg is retained on the endoplasmic reticulum

(PMID: 12609996; PMID: 12221133). We find almost no endogenous Gb1 on the endoplasmic reticulum, and a high density on the plasma membrane. Second, we are able to detect large increases in free HiBit-Gbg after G protein activation using free Gbg sensors (e.g. Fig. 1). Third, many proteins that bind to free Gbg are found entirely in the cytosol of HEK 293 cells

(e.g. PMID: 10066824), suggesting there is not a large population of free Gbg. We have added discussion of these points to the revised manuscript as follows:

“Endogenous Ga and Gb subunits are expressed at approximately a 1:1 ratio, and Gb subunits are tightly associated with Gg and inactive Ga subunits (Cho et al., 2022; Gilman, 1987; Krumins & Gilman, 2006). Moreover, proteins that bind to free Gbg dimers are found in the cytosol of unstimulated HEK 293 cells, suggesting at most only a small population of free Gbg in these cells. Therefore, we assume that the large majority of mNG-b1 and HiBit-b1 subunits in unstimulated cells are part of heterotrimers.”

“Notably, when Gbg dimers are expressed alone they accumulate on the endoplasmic reticulum

(Michaelson et al., 2002; Takida & Wedegaertner, 2003). That we detect almost no endogenous Gbg on the endoplasmic reticulum supports our conclusion that the large majority of Gbg in unstimulated HEK 293 cells is associated with Ga, although we cannot rule out a small population of free Gbg.”

We do not entirely understand the suggested experiment, as free Gbg will still be largely associated with the membrane fraction. Notably, we find almost no HiBit-b1 in the supernatant after lysis in hypotonic buffer and preparation of membrane fractions, and the small amount that we do find does not change if Ga is overexpressed.

Additional context and questions:

(1) There exists some evidence that certain GPCRs can form enduring complexes with Gbetagamma (PubMed: 23297229, 27499021). That would seem to offer a mechanism that would enable receptor-mediated transport of G protein subunits. It would be helpful for the authors to place the findings of this manuscript in the context of these previous findings since they seem somewhat contradictory.

We agree. In our original submission we noted “It is possible that other receptors will influence G protein distribution using mechanisms not shared by the receptors we studied.” In the revised manuscript we have added:

“For example, a few receptors are thought to form relatively stable complexes with Gbg, which could provide a mechanism of trafficking to endosomes (Thomsen et al., 2016; Wehbi et al., 2013).”

(2) There is some evidence that GaS undergoes measurable dissociation from the plasma membrane upon activation (see the mechanism of the assay in PubMed: 35302493). It seems possible that G-alpha (and in particular GaS) might behave differently than the G-beta subunit studied here. This is not entirely clear from the discussion as it now stands.

Indeed, there is abundant evidence that some Gas translocates away from the plasma membrane upon activation. We referred to translocation of “some Ga subunits” in the introduction, although we did not specify that Gas is by far the most studied example. In a previous study (PMID: 27528603) we found that overexpressed Gas samples many intracellular membranes upon activation and returns to the plasma membrane when activation ceases. This is similar to activation-dependent translocation of free Gbg dimers. Because these translocation mechanisms depend on activation and are reversible they are unlikely to be a major source of inactive heterotrimers for intracellular membranes.

We did a poor job of making it clear that we intentionally avoided translocation mechanisms that operate only *during* receptor and G protein stimulation. In the revised manuscript we

have added new data showing reversible activation-dependent translocation of endogenous HiBitGb1.

(3) The authors say "The presence of mNG-b1 on late endosomes suggested that some G proteins may be degraded by lysosomes". The mechanism of lysosomal degradation by proteins on the outside of the lysosome is not clear. It would be helpful for the authors to clarify.

We agree we didn't connect the dots here. Our initial idea was that G proteins on the surface of late endosomes might reach the interior of late endosomes and then lysosomes by involution into multivesicular bodies. However, the reviewer correctly points out that much of the G protein associated with lysosomes still appears to be on the cytosolic surface, where it would not be subject to degradation. In fact, since lysosomes can fuse with the plasma membrane under certain circumstances, this could even represent a pathway for recycling G proteins to the plasma membrane.

We have revised the text to avoid giving the impression that lysosomes degrade G proteins, since we have scant evidence that this occurs. In the revised discussion we point out that we do not know the fate of G proteins located on the surface of lysosomes and speculate that these could be returned to the plasma membrane:

"We do not know the fate of G proteins located on the surface of lysosomes. Since lysosomes may fuse with the plasma membrane under certain circumstances (Xu & Ren, 2015), it is possible that this represents a route of G protein recycling to the plasma membrane."

(4) Although the authors do a good job of assessing G protein dilution in endosomal membranes, it is unclear how this behavior compares to the measurement of other lipidanchored proteins using the same approach. Is the dilution of G proteins what we would expect for any lipid-anchored protein at the inner leaflet of the plasma membrane?

This is a great question. To begin to address it we have studied a model lipid-anchored protein consisting of mNeongreen2 anchored to the plasma membrane by the C terminus of HRas, which is palmitoylated and prenylated. We find that this protein is also diluted on endocytic vesicles, although to a lesser degree than heterotrimeric G proteins. We have added a section to the results and a new figure supplement describing these results:

"To test if other peripheral membrane proteins are similarly depleted from endocytic vesicles, we performed analogous experiments by overexpressing mNG bearing the C-terminal membrane anchor of HRas (mNG-HRas ct). We found that mNG-HRas ct was also less abundant on FM464-positive endocytic vesicles than expected based on plasma membrane abundance, although not to the same extent as mNG-b1 (Figure 4 - figure supplement 2); mNG-HRas ct density on FM4-64-positive vesicles was $64 \pm 17\%$ (mean $\pm$ 95% CI; n=78) of the nearby plasma membrane."

Reviewer #2 (Public Review):

This is an interesting method that addresses the important problem of assessing G protein localization at endogenous levels. The data are generally convincing.

Specific comments

Methods:

The description of the gene editing method is unclear. There are two different CRISPR cell lines made in two different cell backgrounds. The methods should clearly state which

CRISPR guides were used on which cell line. It is also not clear why HiBit is included in the mNG-β1 construct. Presumably, this is not critical but it would be helpful to explicitly note. In general, the Methods could be more complete.

We have added the following to the methods to clarify that the same gRNA was used to produce both cell lines:

“The human GNB1 gene was targeted at a site corresponding to the N-terminus of the Gb1 protein; the sequence 5'-TGAGTGAGCTTGACCAGTTA-3' was incorporated into the crRNA, and the same gRNA was used to produce both HiBit-b1 and mNG-b1 cell lines.”

We have added the following to the methods to clarify why HiBit is included in the mNG-b1 construct:

“HiBit was included in the repair template for producing mNG-b1 cells to enable screening for edited clones using luminescence.”

Results:

The explanation of validation experiments in Figures 1 C and D is incomplete and difficult to follow. The rationale and explanation of the experiments could be expanded. In addition, because this is an interesting method, it would be helpful to know if the endogenous editing affects normal GPCR signaling. For example, the authors could include data showing an Isoinduced cAMP response. This is not critical to the present interpretation but is relevant as a general point regarding the method. Also, it may be relevant to the interpretation of receptor effects on G protein localization.

We have expanded the rationale and explanation of experiments in Figures 1C and D by adding:

“For example, we observed agonist-induced BRET between the D2 dopamine receptor and mNG-b1, an interaction that requires association with endogenous Ga subunits (Figure 1C). Similarly, we observed BRET between HiBit-b1 and the free Gbg sensor memGRKct-Venus after activation of receptors that couple Gi/o, Gs, and Gq heterotrimers, indicating that HiBit-b1 associated with endogenous Ga subunits from these three families (Figure 1D).”

We have done the suggested cAMP experiment and provide the data in a new figure supplement:

“We also found that cyclic AMP accumulation in response to stimulation of endogenous β adrenergic receptors was similar in edited cell lines and their unedited parent lines (Figure 1 - figure supplement 1).”

Discussion:

The conclusion that beta-gamma subunits do not redistribute after GPCR activation seems new and different from previous reports. Is this correct? Can the authors elaborate on how the results compare to previous literature?

Many previous studies have indeed shown that *free* Gbg dimers can redistribute after GPCR activation and sample intracellular membranes. Our initial focus was on possible changes in heterotrimer distribution after GPCR activation, but in retrospect we should have directly addressed free Gbg translocation and made the distinction clear.

In the revised manuscript we show that *during* stimulation we observe changes consistent with modest translocation of endogenous Gbg from the plasma membrane and sampling of

intracellular compartments. To our knowledge this is the first demonstration of endogenous Gbg translocation.

We have added:

“With overexpressed G proteins free Gbg dimers translocate from the plasma membrane and sample intracellular membrane compartments after activation-induced dissociation from Ga subunits. Consistent with this, we observed small decreases in bystander BRET at the plasma membrane and small increases in bystander BRET at intracellular compartments during activation of GPCRs, suggesting that endogenous Gbg subunits undergo similar translocation (Figure 5- figure supplement 1). Notably, these changes occurred at room temperature, suggesting that endocytosis was not involved, and developed over the course of minutes. The latter observation and the small magnitude of agonist-induced changes are both consistent with expression of primarily slowly-translocating endogenous Gg subtypes in HEK 293 cells. Moreover, as shown previously for overexpressed Gbg, the changes we observed with endogenous Gbg were readily reversible (Figure 5- figure supplement 1), suggesting that most heterotrimers reassemble at the plasma membrane after activation ceases.”

Can the authors note that OpenCell has endogenously tagged Gβ1 and reports more obvious internal localization? Can the authors comment on this point?

OpenCell has tagged GNB1 and the Leonetti group kindly provided a parent cell line we used to add a slightly different tag. Although their study did not identify any specific intracellular compartments, our impression is that most of the internal structures visible in their images are likely to be lysosomes, as they are large, round and often have a clear lumen. Overall their images and ours are comfortably similar. We have added:

“Unsurprisingly, our images are quite similar to those made as part of previous study that labeled Gb1 subunits with mNG2 (Cho et al., 2022).”

Notably, the Leonetti group has recently reported the subcellular distribution of many untagged proteins using a proteomic approach. They find that Gb1 is enriched on the plasma membrane and lysosomes but is not enriched on endosomes, the Golgi apparatus, endoplasmic reticulum or mitochondria (<https://www.biorxiv.org/content/10.1101/2023.12.18.572249v1>). We have cited this work in the revised manuscript.

Is this the first use of CRISPR / HiBit for BRET assay? It would be helpful to know this or cite previous work if not. Also, as this is submitted as a tools piece, the authors might say a little more about the potential application to other questions.

The only previous study we are aware of utilizing a similar combination of methods is a 2020 report from the group of Dr. Stephen Hill, in which the authors studied binding of fluorescent ligands to HiBit-tagged GPCRs. This work is now cited.

We have also added the following to our previous brief statement about potential applications:

“In addition, it may also be possible to use these cells in combination with targeted sensors to study endogenous G protein activation in different subcellular compartments. More broadly, our results show that subcellular localization of endogenous membrane proteins can be studied in living cells by adding a HiBit tag and performing bystander BRET mapping. Applied at large scale this approach would have some advantages over fluorescent protein complementation, most notably the ability to localize endogenous membrane proteins that are expressed at levels that are too low to permit fluorescence microscopy.”

Reviewer #3 (Public Review):

Summary:

This article addresses an important and interesting question concerning intracellular localization and dynamics of endogenous G proteins. The fate and trafficking of G protein-coupled receptors (GPCRs) have been extensively studied but so far little is known about the trafficking routes of their partner G proteins that are known to dissociate from their respective receptors upon activation of the signaling pathway. The authors utilize modern cell biology tools including genome editing and bystander bioluminescence resonance energy transfer (BRET) to probe intracellular localization of G proteins in various membrane compartments in steady state and also upon receptor activation. Data presented in this manuscript shows that while G proteins are mostly present on the plasma membrane, they can be also detected in endosomal compartments, especially in late endosomes and lysosomes. This distribution, according to data presented in this study, seems not to be affected by receptor activation. These findings will have implications in further studies addressing GPCR signaling mechanisms from intracellular compartments.

Strengths:

The methods used in this study are adequate for the question asked. Especially, the use of genome-edited cells (for the addition of the tag on one of the G proteins) is a great choice to prevent the effects of overexpression. Moreover, the use of bystander BRET allowed authors to probe the intracellular localization of G proteins in a very high-throughput fashion. By combining imaging and BRET authors convincingly show that G proteins are very low abundant on early endosomes (also ER, mitochondria, and medial Golgi), however seem to accumulate on membranes of late endosomal compartments.

Weaknesses:

While the authors provide a novel dataset, many questions regarding G protein trafficking remain open. For example, it is not entirely clear which pathway is utilized to traffic G proteins from the plasma membrane to intracellular compartments. Additionally, future studies should also address the dynamics of G protein trafficking, for example by tracking them over multiple time points.

We agree, there is much more to do.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

On page 7 the text says "the difference did reach significance (Figure 5D)". It looks like the difference did not reach significance. Please check on this.

Thank you, this was an unfortunately significant typo.

Reviewer #3 (Recommendations For The Authors):

This article addresses an important and interesting question concerning intracellular localization and dynamics of endogenous G proteins. While the posed question is indeed a grand one and the methods used by the authors are novel, I believe that the data presented in this manuscript are still insufficient to support all claims posed by the authors. Below I list my major concerns:

(1) The authors claim that they provide a "detailed subcellular map of endogenous G protein distribution", however, the map is in my opinion not sufficiently detailed (e.g. trans-Golgi network is not included) and not quantitative enough (e.g. % of proteins present on one compartment vs. the other as authors claim that BRET signals "cannot be directly compared between different compartments"). To strengthen this statement, except for providing more extensive and quantitative data, it would be beneficial to provide such a "map" as an illustration based on the findings presented in this article.

“Detailed” is certainly a subjective term. While we maintain that our description of endogenous G protein distribution is far more detailed than any previous study, we now simply claim to provide a “subcellular map”. We have added images of TGNP (TGN46; TGOLN2), showing that endogenous G proteins are readily detectable on the structures labeled by this marker. These data are now provided in Figure 3 – figure supplement 7.

We did not claim that our study was quantitative- we did not try to count G proteins. However, if we use published estimates of total G proteins and surface area for HEK 293 cells we estimate that there are roughly 2,500 G proteins μm^{-2} on the plasma membrane and 500 G proteins μm^{-2} on endocytic vesicles. For other intracellular compartments relative density can be approximated by inspecting images, but a truly quantitative estimate would require a surface area standard analogous to FM4-64 for each compartment. The percentage of the total G protein pool on a given compartment is, in our opinion, less important than the density of G proteins on that compartment, as the latter is more likely to affect the efficiency of local signal transduction. Since we do not claim to have accurate G protein density estimates for many intracellular compartments, we prefer to provide several raw images for each compartment rather than a schematized map.

Bystander BRET values cannot be compared directly across compartments due to differences in expression and energy transfer efficiency of different markers and compartment surface area. This method is well suited for following changes in distribution as a function of time or after perturbations and for sensitive detection of weak colocalization but can only provide approximate “maps” of absolute distribution.

(2) Probing of the intracellular distribution of these proteins, especially after GPCR activation, includes a single chosen timepoint. I believe that the manuscript would greatly benefit from including some dynamic data on internalization and intracellular trafficking kinetics. What is the turnover of tested G proteins? What is the fraction that is going to recycling compartments and/or lysosomes? Authors could perhaps turn to other methods to be able to dynamically track proteins over time e.g. via photoconversion techniques.

Because G protein trafficking appears to be largely constitutive there is no easy way for us to assess how long it takes G proteins to transit various intracellular compartments, although we agree this would be interesting. As the reviewer suggests, dynamic data on constitutive trafficking would require methods (such as photoconversion) not currently available to us for endogenous G proteins. Accordingly, we have made no claims regarding the kinetics of G protein trafficking. As for possible redistribution after GPCR activation, in the revised manuscript we have added 5- and 15-minute timepoints after agonist stimulation for our bystander BRET mapping (Figure 5- figure supplement 2). These timepoints were chosen to correspond to persistent signaling mediated by internalized receptors.

(3) Exemplary images with cells showing significant colocalization with lysosomal compartments seem to contain more intracellular vesicles visible in the mNG channel than in the case of the other compartment. Is it an effect of the treatment to stain

lysosomes? It would be beneficial to compare it with some endogenous marker e.g. LAMP1 without additional treatments.

The visibility of intracellular vesicles in our lysosome images likely reflects our selection of cells and regions with visible and abundant lysosomes, specifically peripheral regions directly adhered to the coverslip, rather than treatment with lysosomal stains (LV 633 and dextran). As suggested, we now include images of cells expressing LAMP1 as an alternative lysosome marker (Figure 3 - figure supplement 6).

(4) The authors probe an abundance of G proteins along the constitutive endocytic pathway. However, to prove that G proteins are not de-palmitoylated rather than endocytosed authors should perform control experiments where endocytosis is blocked e.g. pharmacologically or via a knockdown approach. Additionally, various endocytic pathways can be probed.

We did not claim that depalmitoylation plays no role in delivery of G proteins to internal compartments. In fact, we pointed out that we cannot at present rule out other pathways and delivery mechanisms. Importantly, if some of the G proteins that we detect along the endocytic pathway do arrive there by trafficking through the cytosol this would only strengthen our major conclusion that endocytosis is inefficient.

Having said this, we have now conducted extensive experiments investigating the role of palmitate cycling in the trafficking of heterotrimeric G proteins and the small G protein H-Ras. Our results suggest that a depalmitoylation-repalmitoylation cycle is not important for the distribution of heterotrimers, but these findings will be the subject of a separate publication focused on this specific question for both large and small G proteins.

We agree that it will be interesting to probe different endocytic pathways, as suggested using a genetic approach. Our main interest here was in endocytic membranes that were defined functionally (with FM4-64 or internalized receptors) rather than biochemically.

Minor comments:

(5) "Imaging" paragraph in the Methods section refers to a non-existent figure called "SI Appendix S9".

Thank you.

(6) It is not clear what was used as a "control" in Figure 5E.

"Control" refers to DPBS vehicle alone. This information is now added to the legend for Figure 5E.

<https://doi.org/10.7554/eLife.97033.2.sa0>